

Isolation Flow

Course Workbook

Table of Contents

About this Workbook

[Slide 3](#)

Pre-Lab: Workshop Pre-requisites

[Slide 4](#)

Lab 3: Xilinx Isolation Flow

[Slide 7](#)

About this Workbook

The contents of this workbook are created by Aduvo Engineering & Training, Ltd.

If you have any questions about the contents, or need assistance, please contact Adam Taylor at adam@aduvoengineering.com.

Pre-Lab

Workshop Pre-requisites

Required Hardware

There is no required hardware for this Lab.

Downloads and Installations

Step 1 – Download and install the following at least one day prior to the workshop. This may take a significant amount of time and drive space.

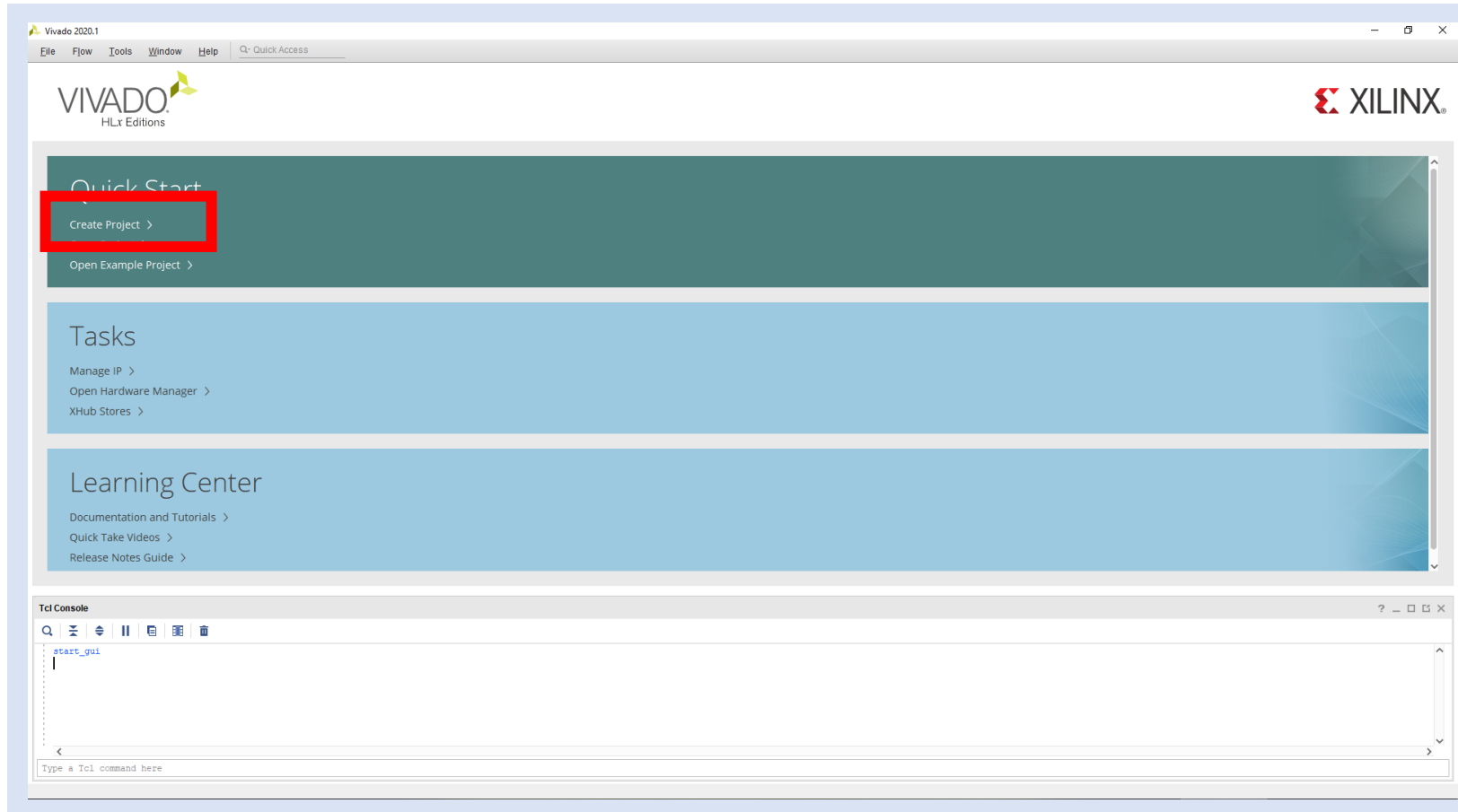
Vivado 2020.1	Download
---------------	--------------------------

Lab 3

Xilinx Isolation Flow

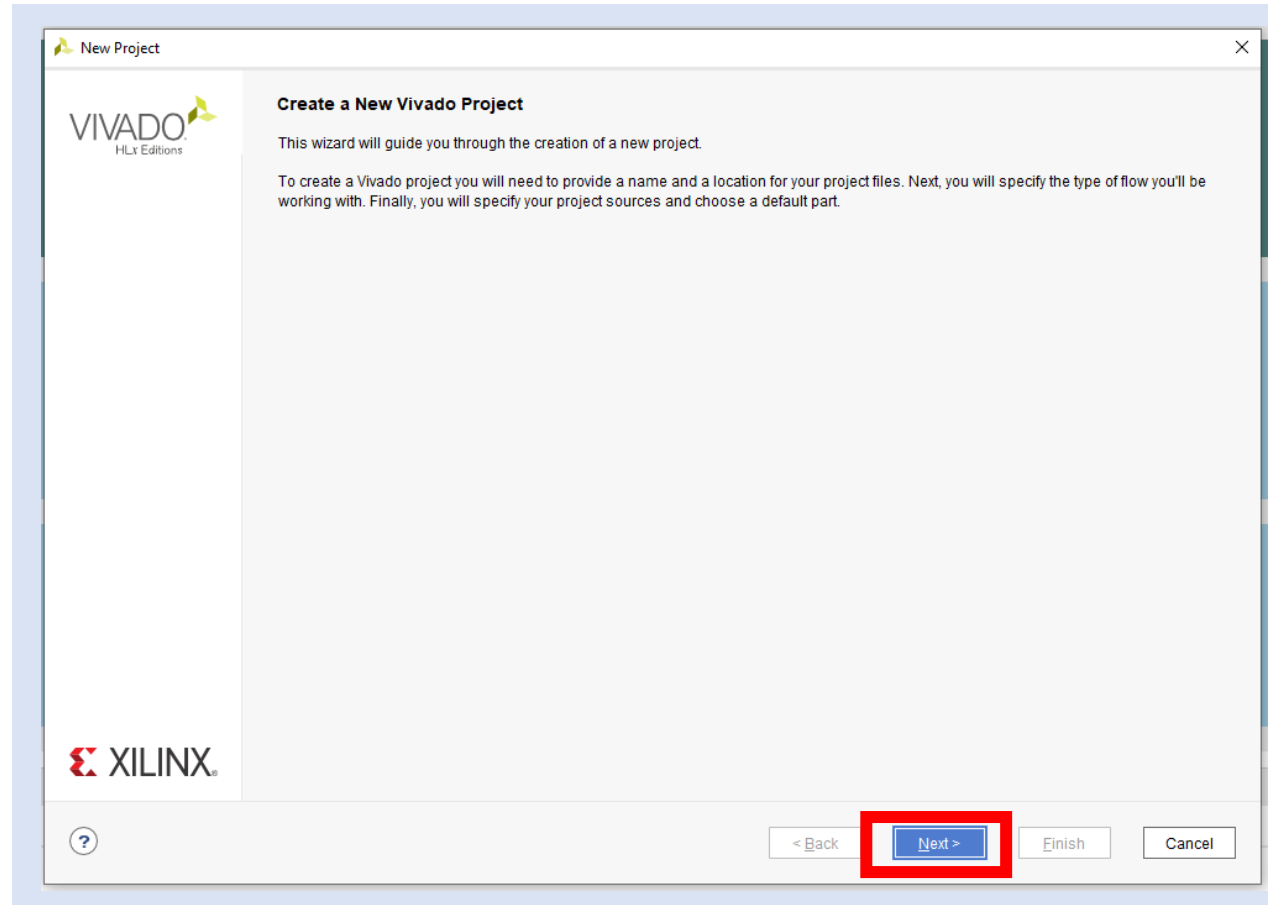
Lab 3: Overview and Introduction to Vivado

Step 1 – Create a new Vivado project



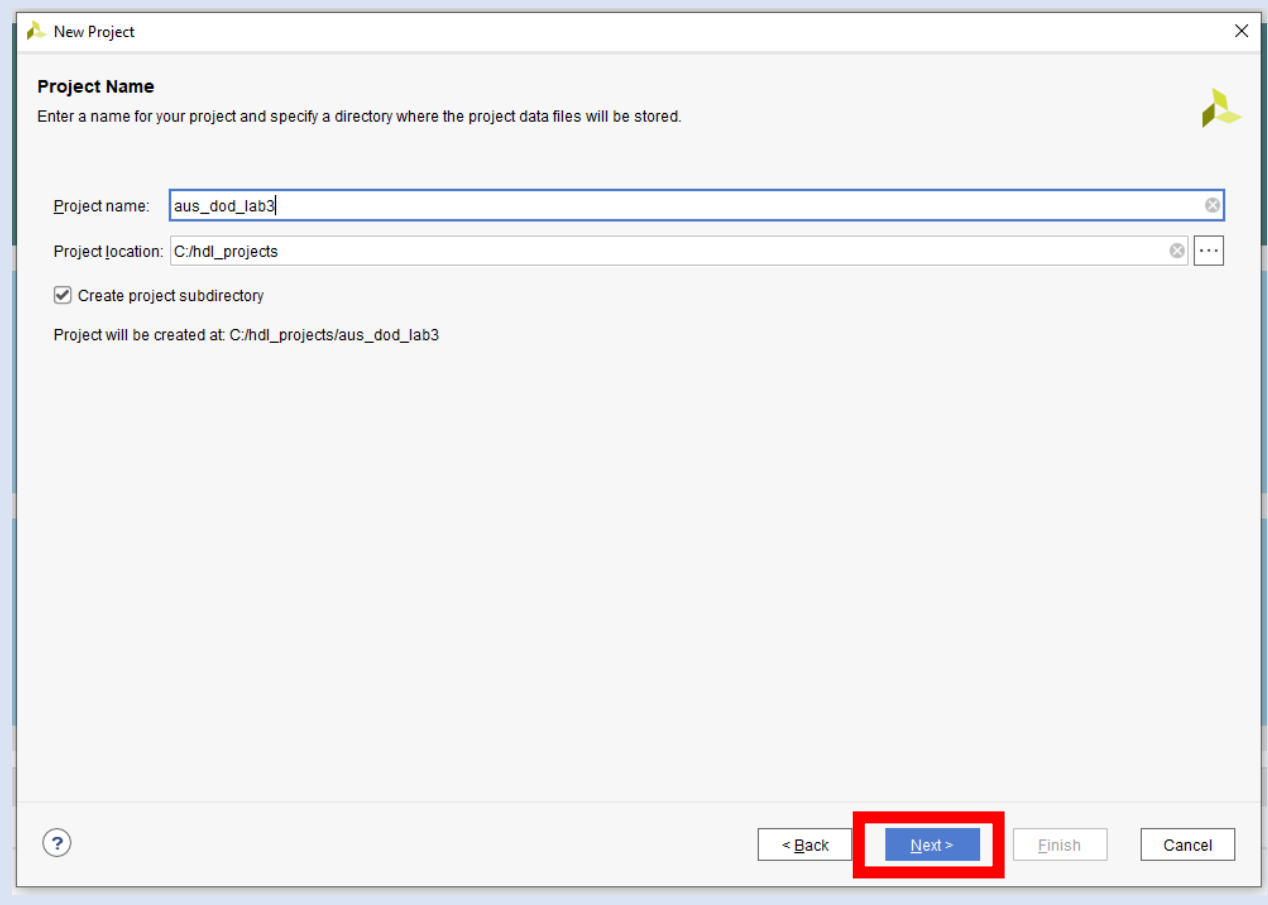
Lab 3: Overview and Introduction to Vivado

Step 2 – Click on next



Lab 3: Overview and Introduction to Vivado

Step 3 – Enter a name and project location



The image shows the 'New Project' dialog box in Vivado. The 'Project Name' field is filled with 'aus_dod_lab3'. The 'Project location' field is filled with 'C:/hdl_projects'. The 'Create project subdirectory' checkbox is checked. The project will be created at 'C:/hdl_projects/aus_dod_lab3'. The 'Next >' button is highlighted with a red rectangle.

New Project

Enter a name for your project and specify a directory where the project data files will be stored.

Project name:

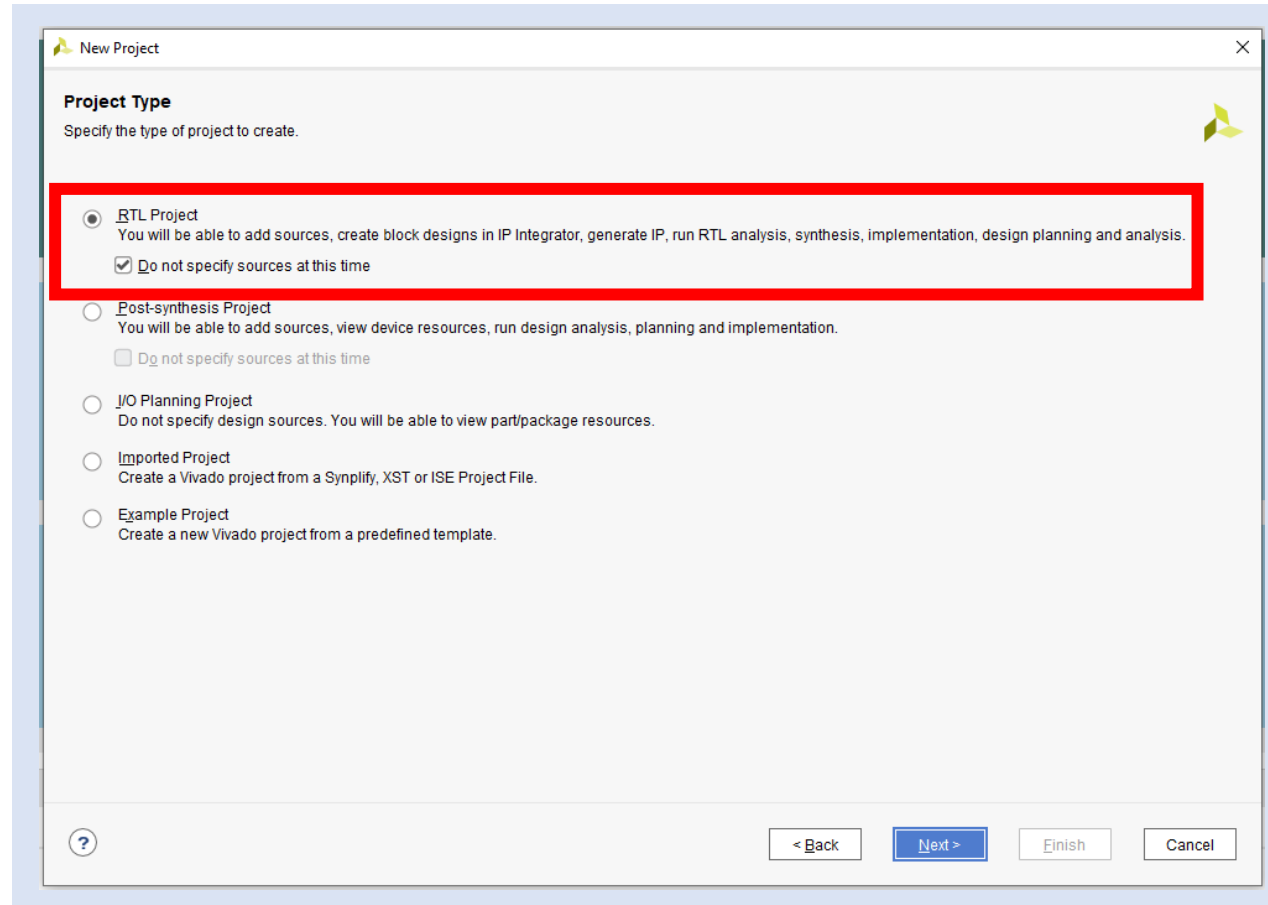
Project location:

☒ Create project subdirectory

Project will be created at: C:/hdl_projects/aus_dod_lab3

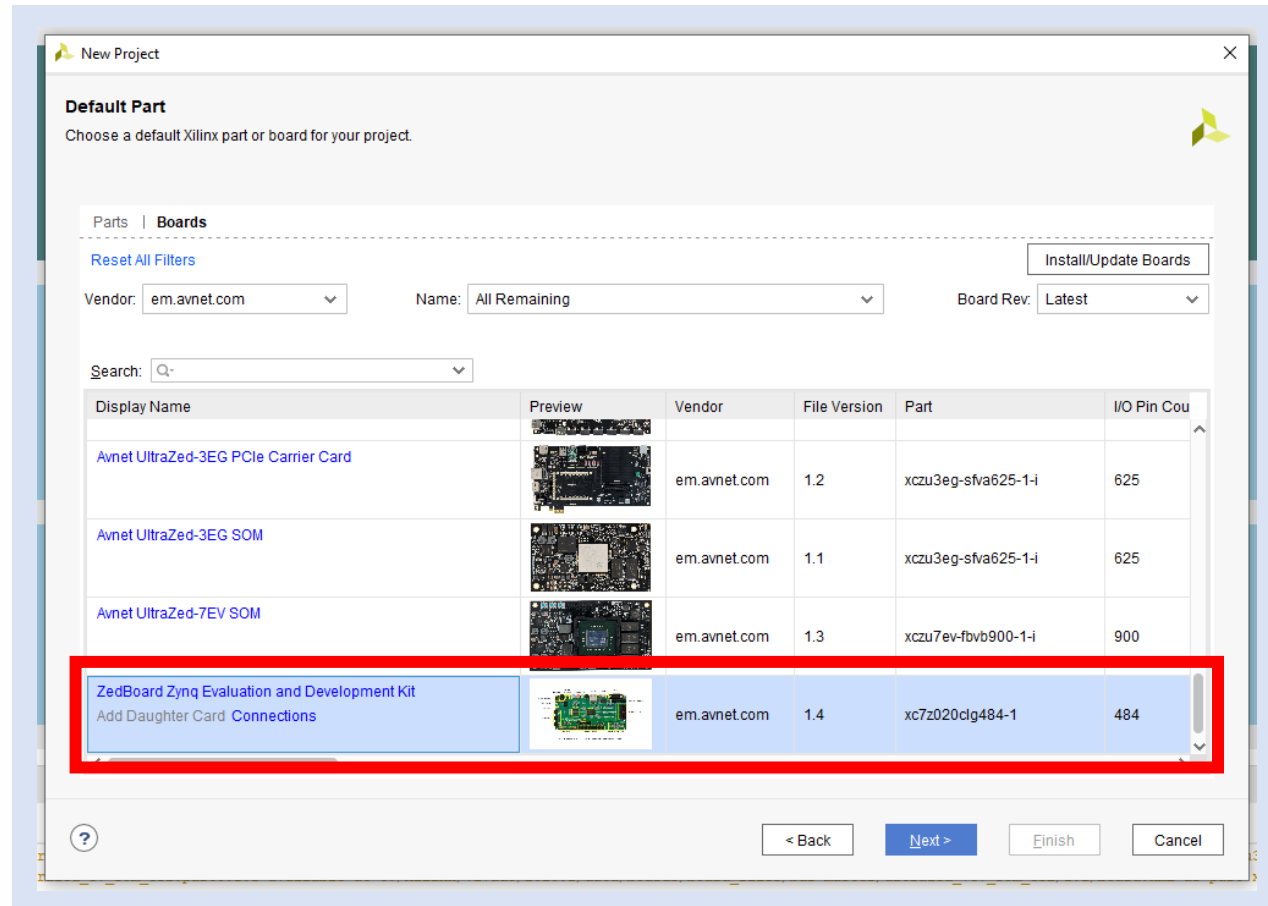
Lab 3: Overview and Introduction to Vivado

Step 4 – Select RTL Project and Check so not specify source at this time



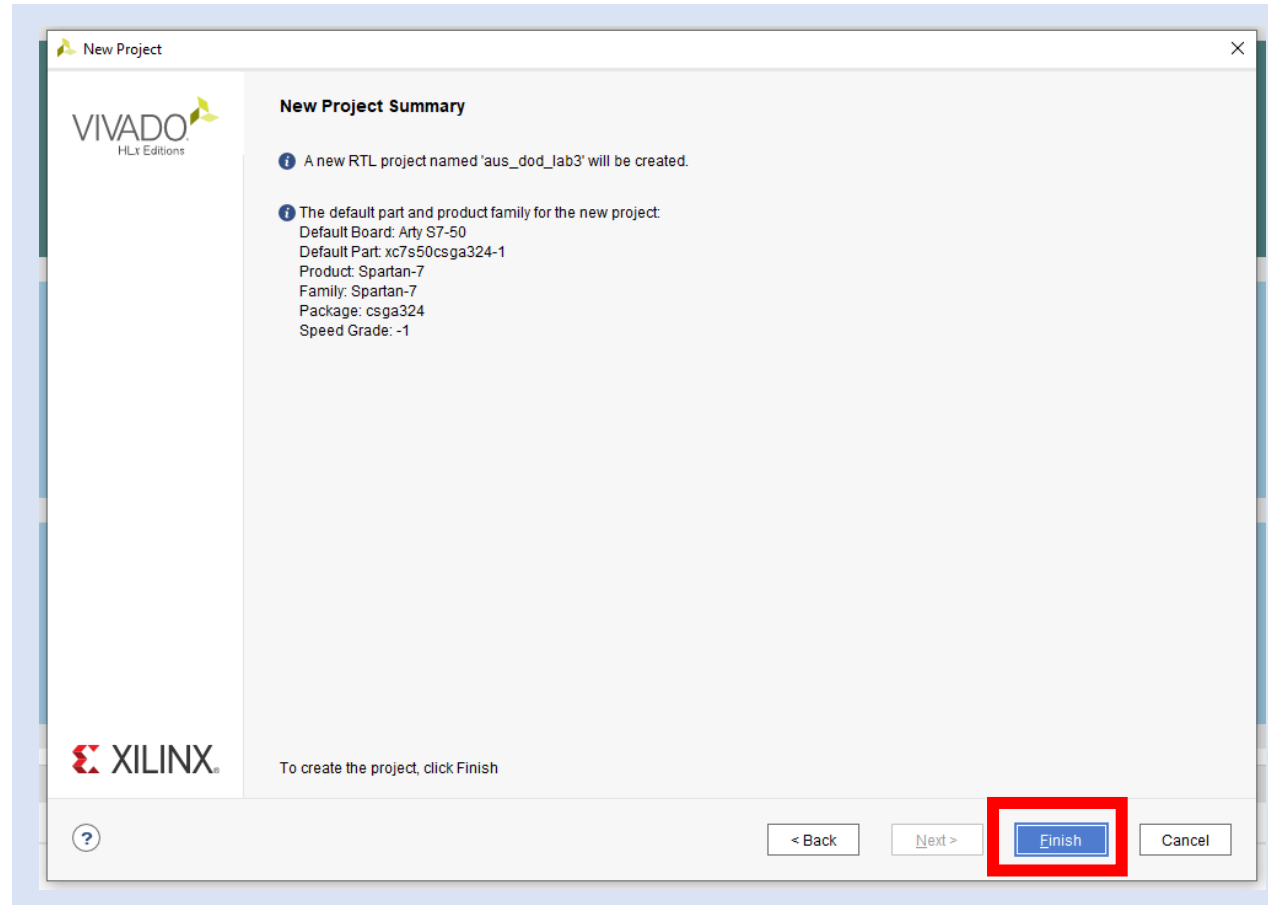
Lab 3: Overview and Introduction to Vivado

Step 5 – Select ZedBoard



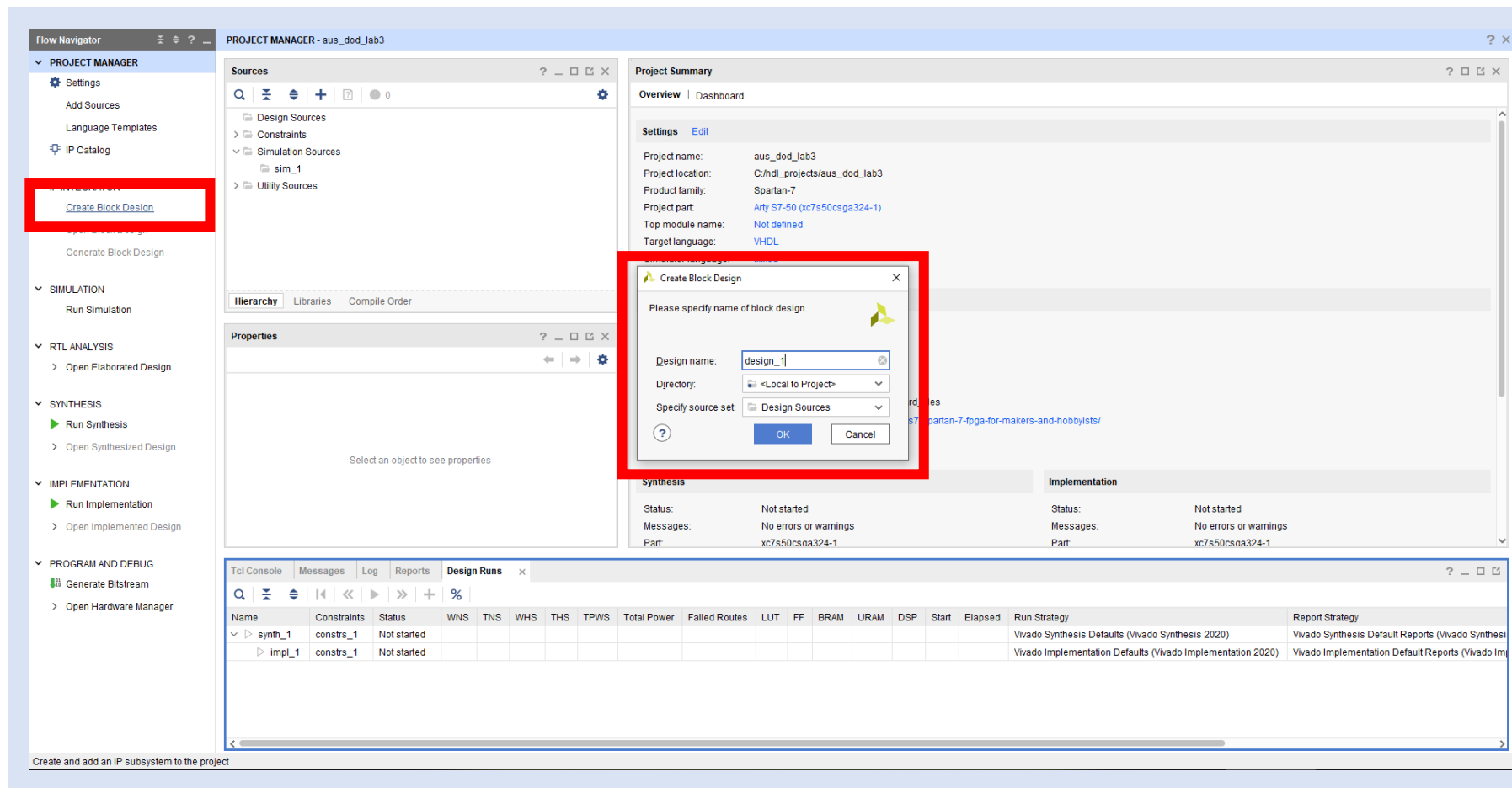
Lab 3: Overview and Introduction to Vivado

Step 6 – Click on Finish



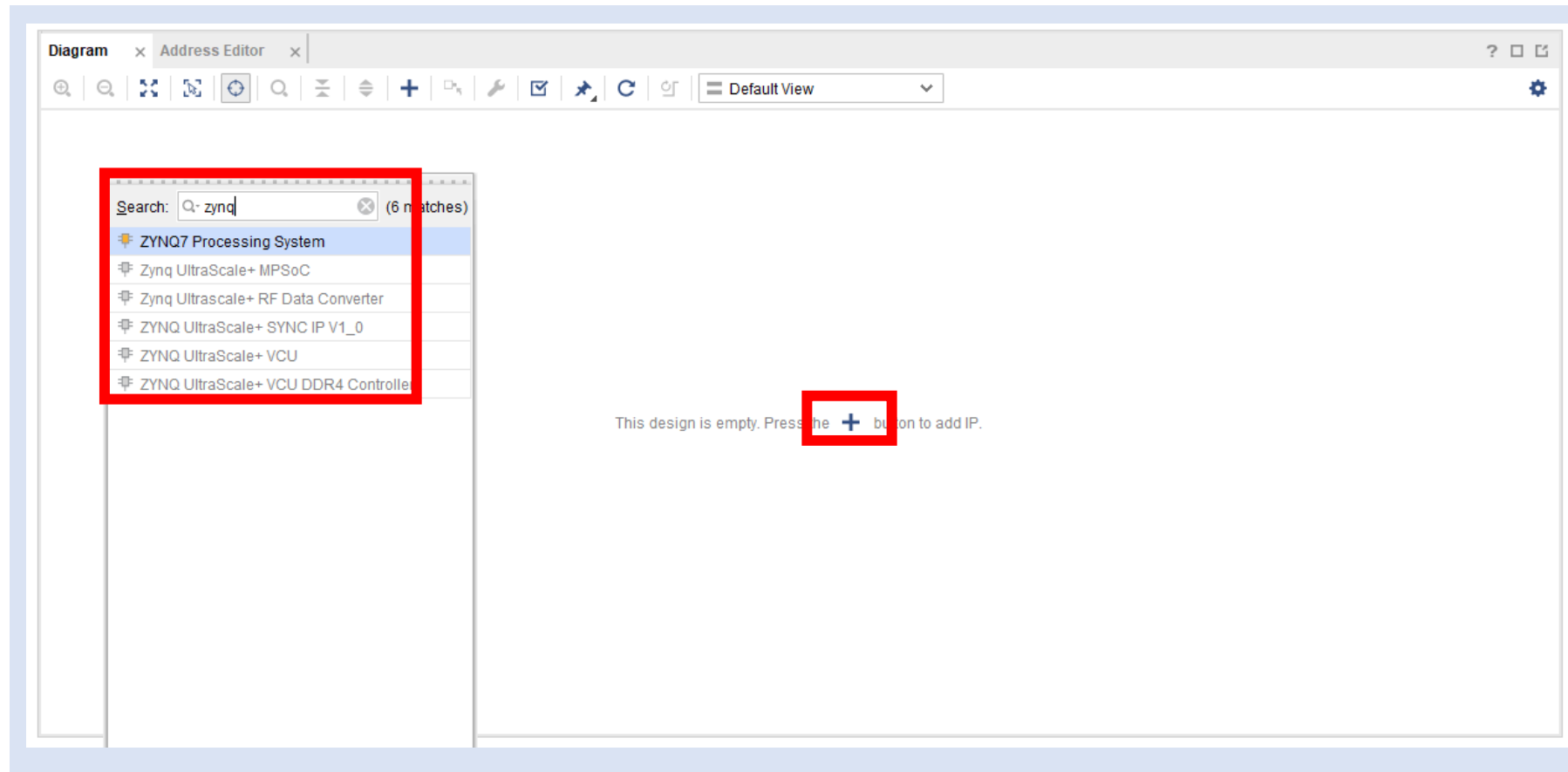
Lab 3: Overview and Introduction to Vivado

Step 7 – Click Create Block Design – leave the dialog unchanged and click OK



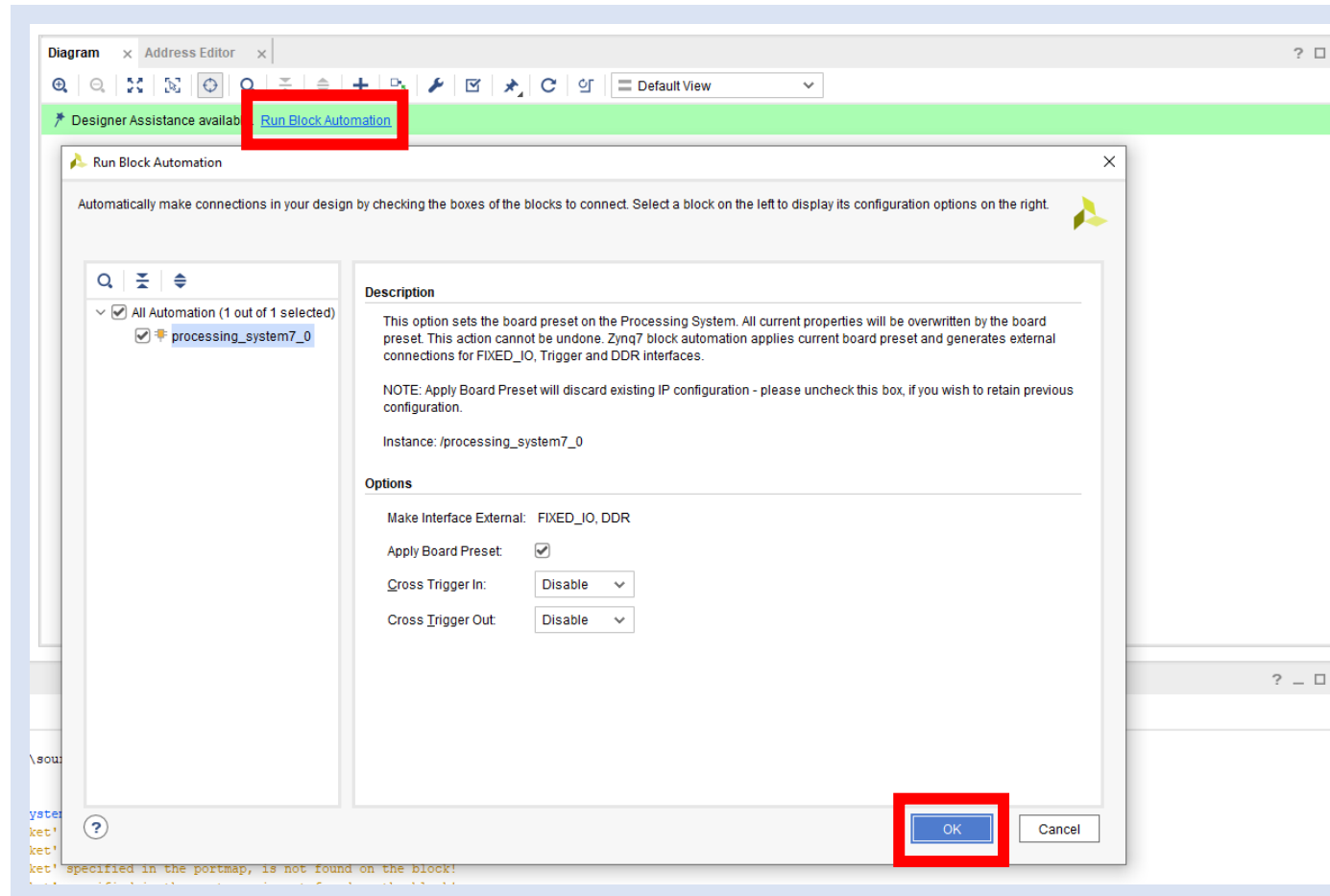
Lab 3: Overview and Introduction to Vivado

Step 8 – Click on the + and type in the search box Zynq, add the ZYNQ PS to the system



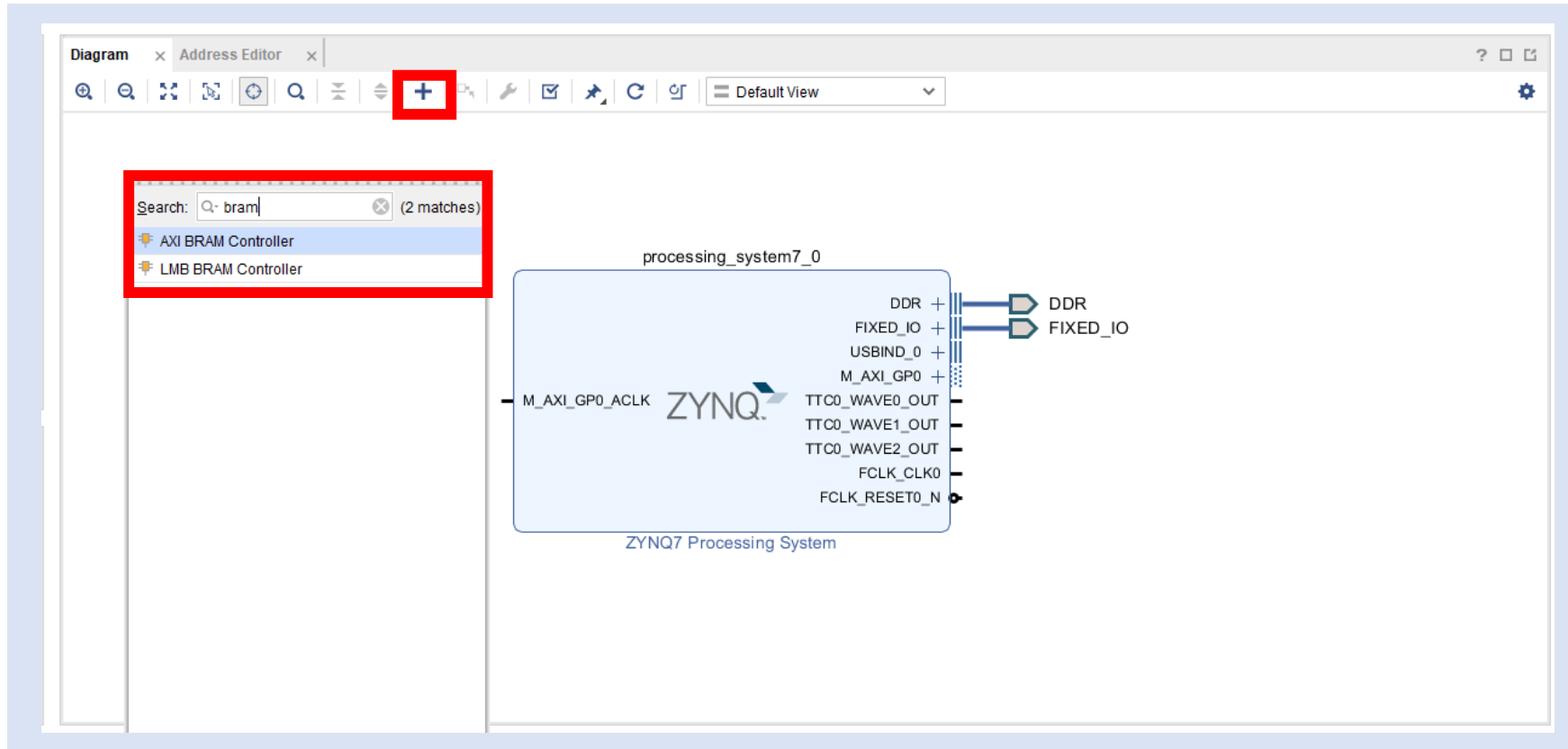
Lab 3: Overview and Introduction to Vivado

Step 9 – Click on Run Block Automation and then click OK



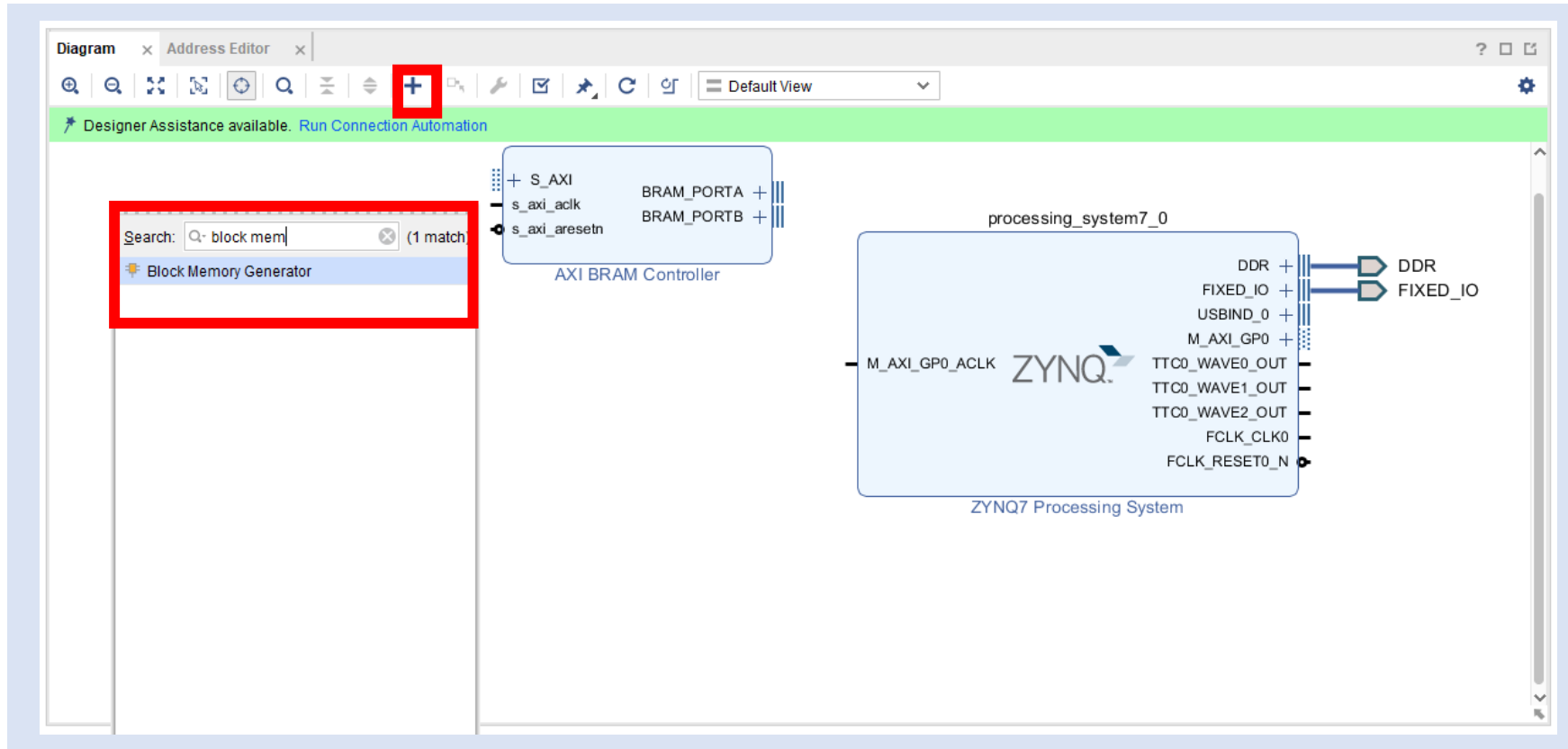
Lab 3: Overview and Introduction to Vivado

Step 10 – Click on + and type in BRAM – Add in the AXI-BRAM Controller



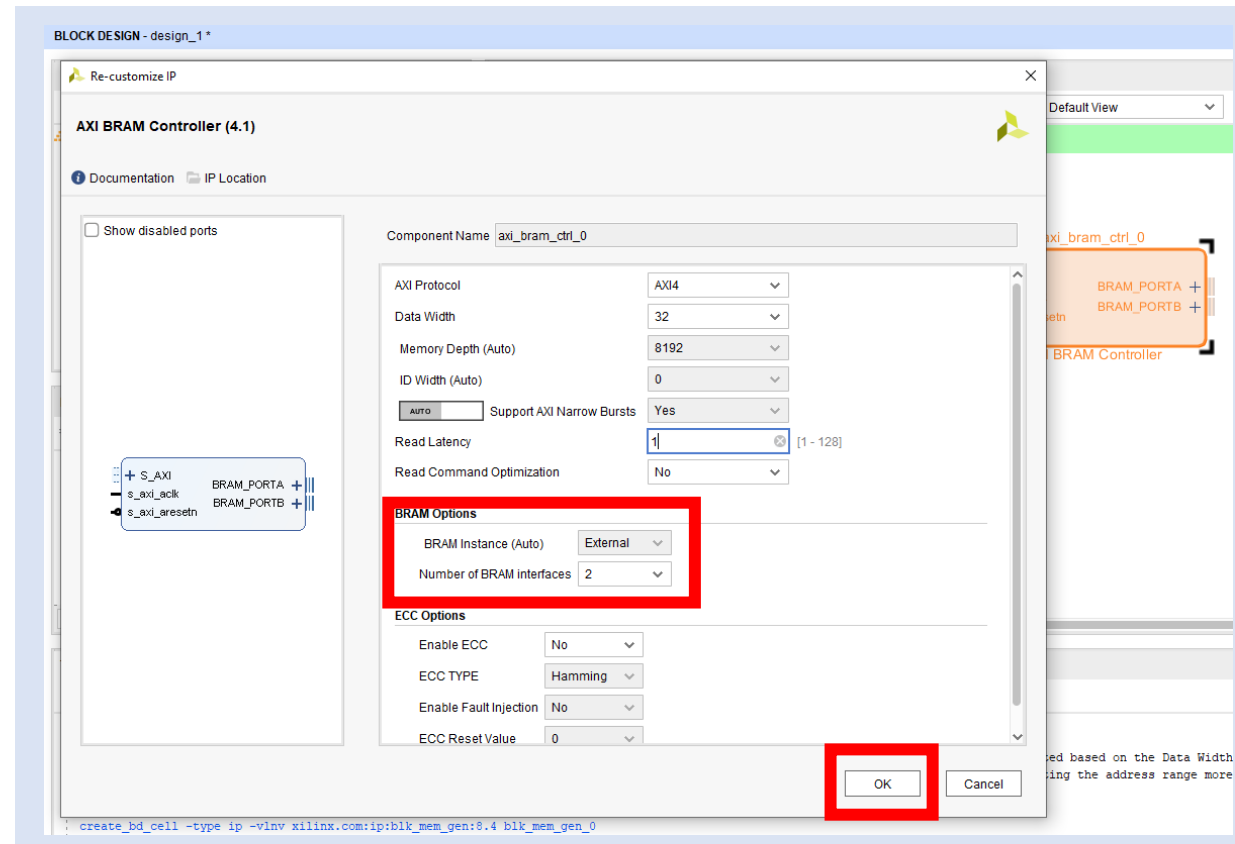
Lab 3: Overview and Introduction to Vivado

Step 11 – Click on + and type in block memory and add in a block memory generator



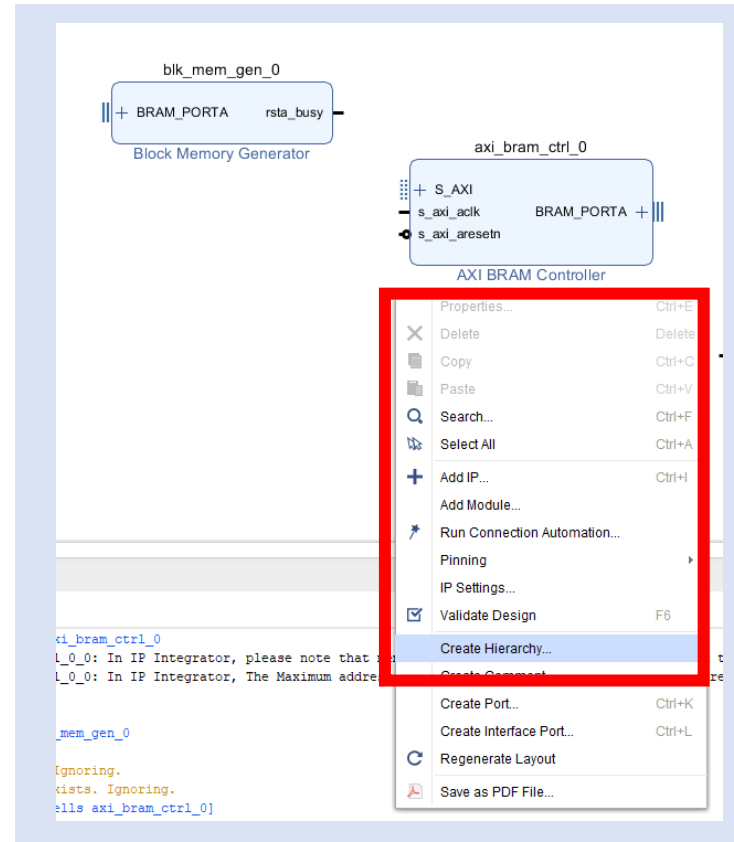
Lab 3: Overview and Introduction to Vivado

Step 12 – Double click on the AXI BRAM controller and change the number of BRAM interfaces to ONE



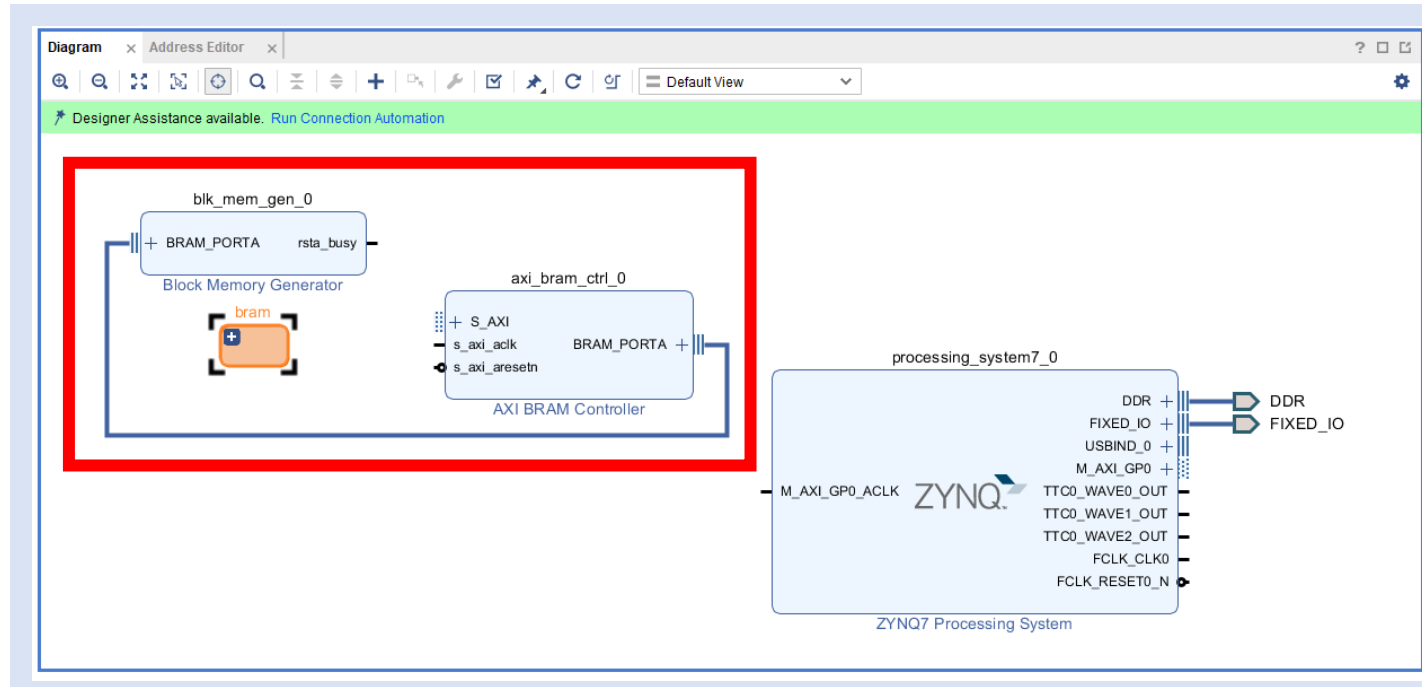
Lab 3: Overview and Introduction to Vivado

Step 13 – Right click anywhere in the block diagram and select create hierarchy – Name it BRAM when prompted



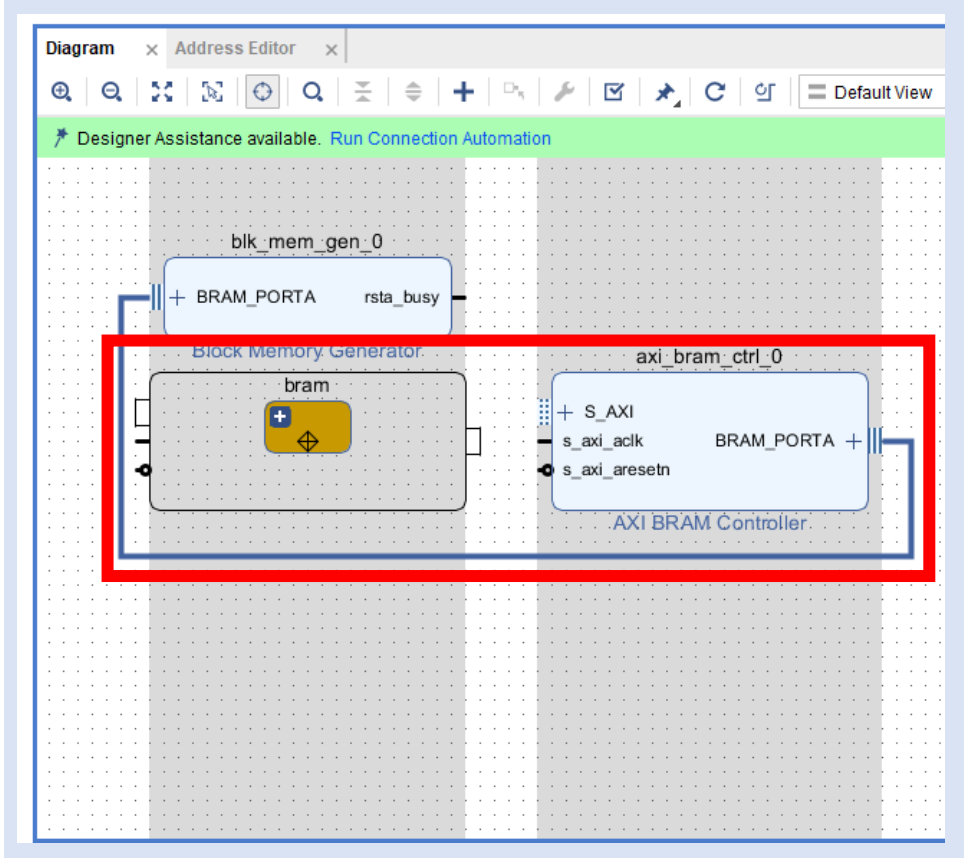
Lab 3: Overview and Introduction to Vivado

Step 14 – Connect the BRAM Controller to BRAM



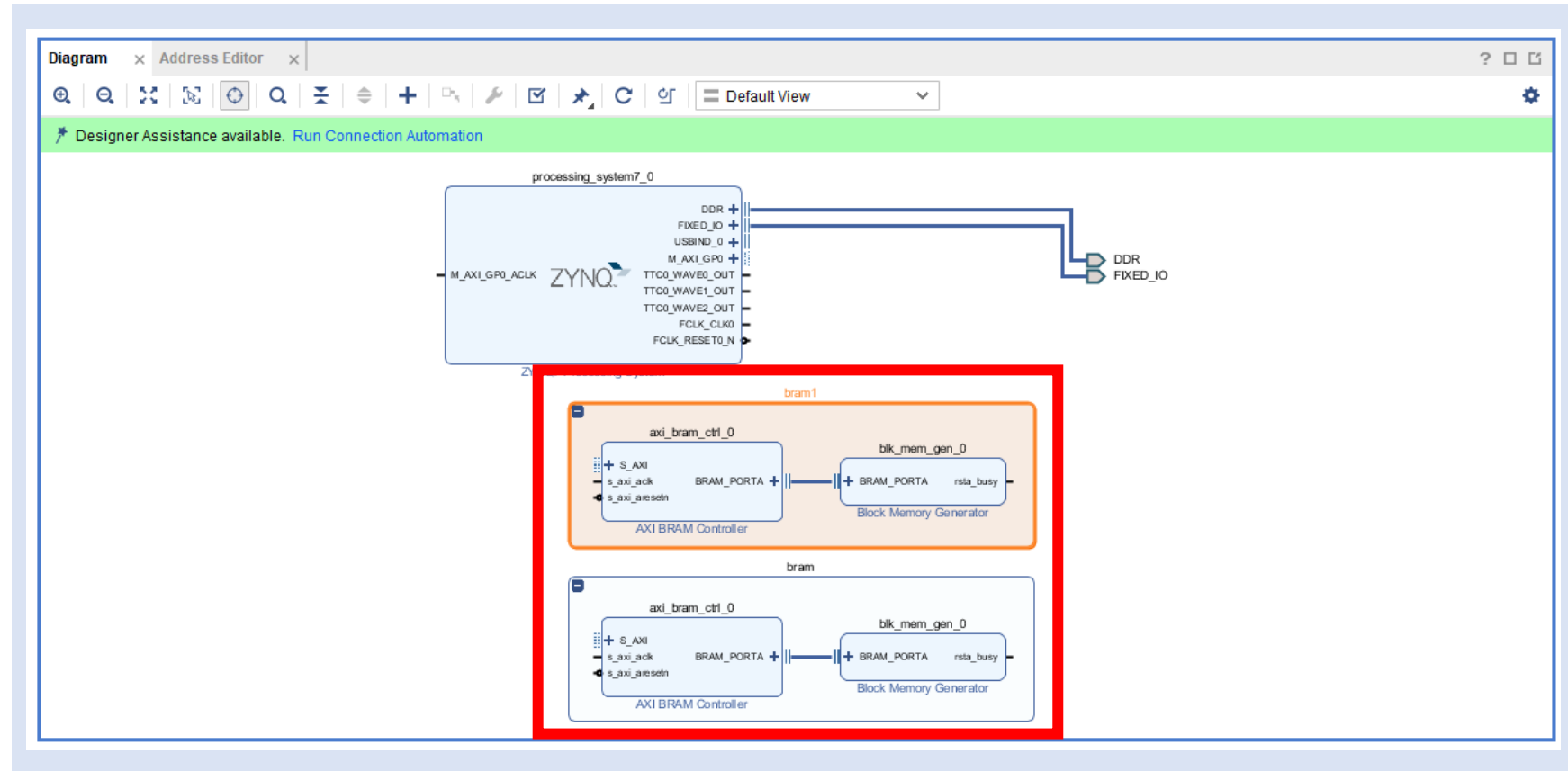
Lab 3: Overview and Introduction to Vivado

Step 15 – Add the BRAM and BRAM Controller to the hierarchy block by dragging and dropping



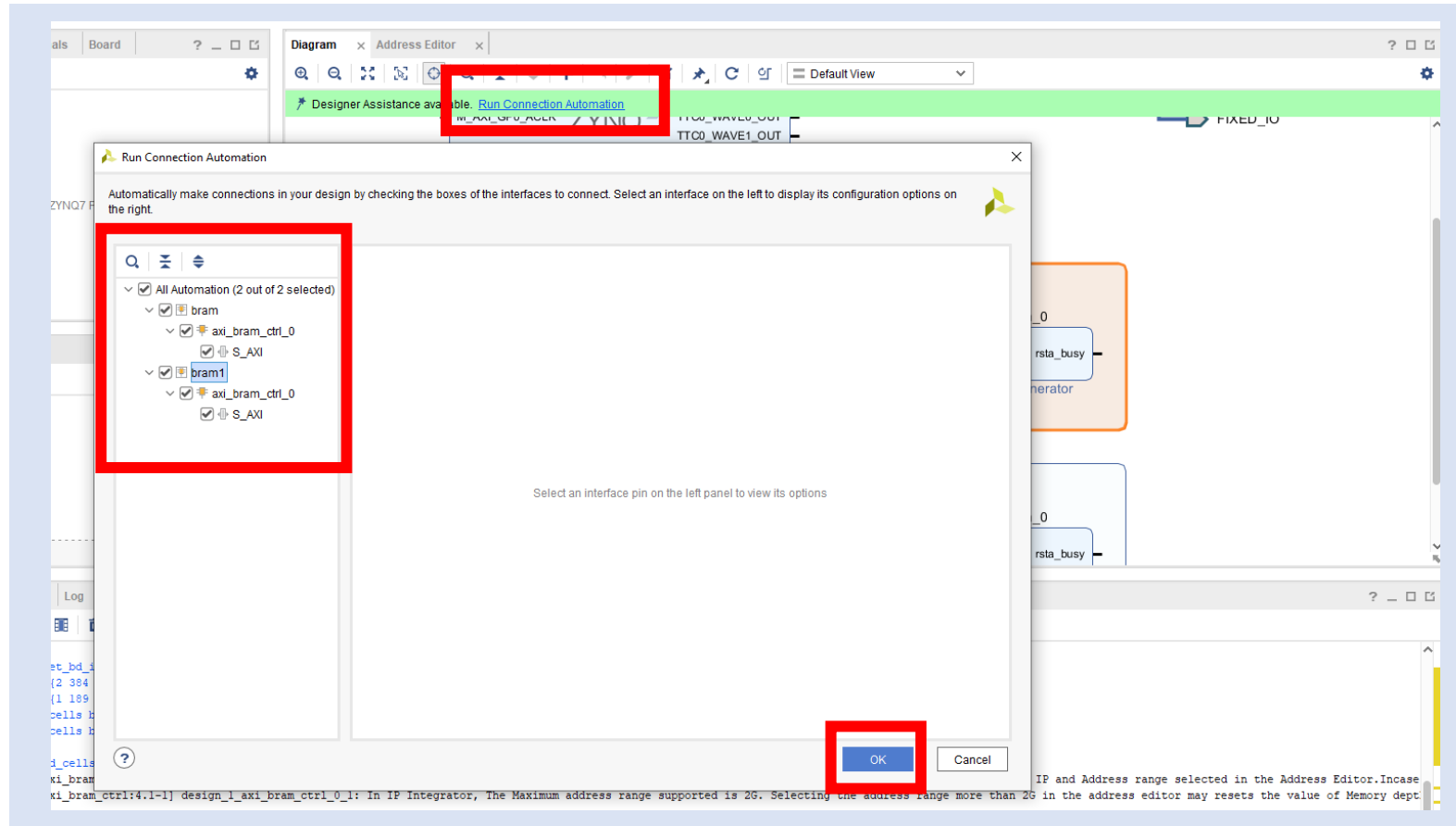
Lab 3: Overview and Introduction to Vivado

Step 16 – Copy and Paste the hierarchy block onto the block diagram



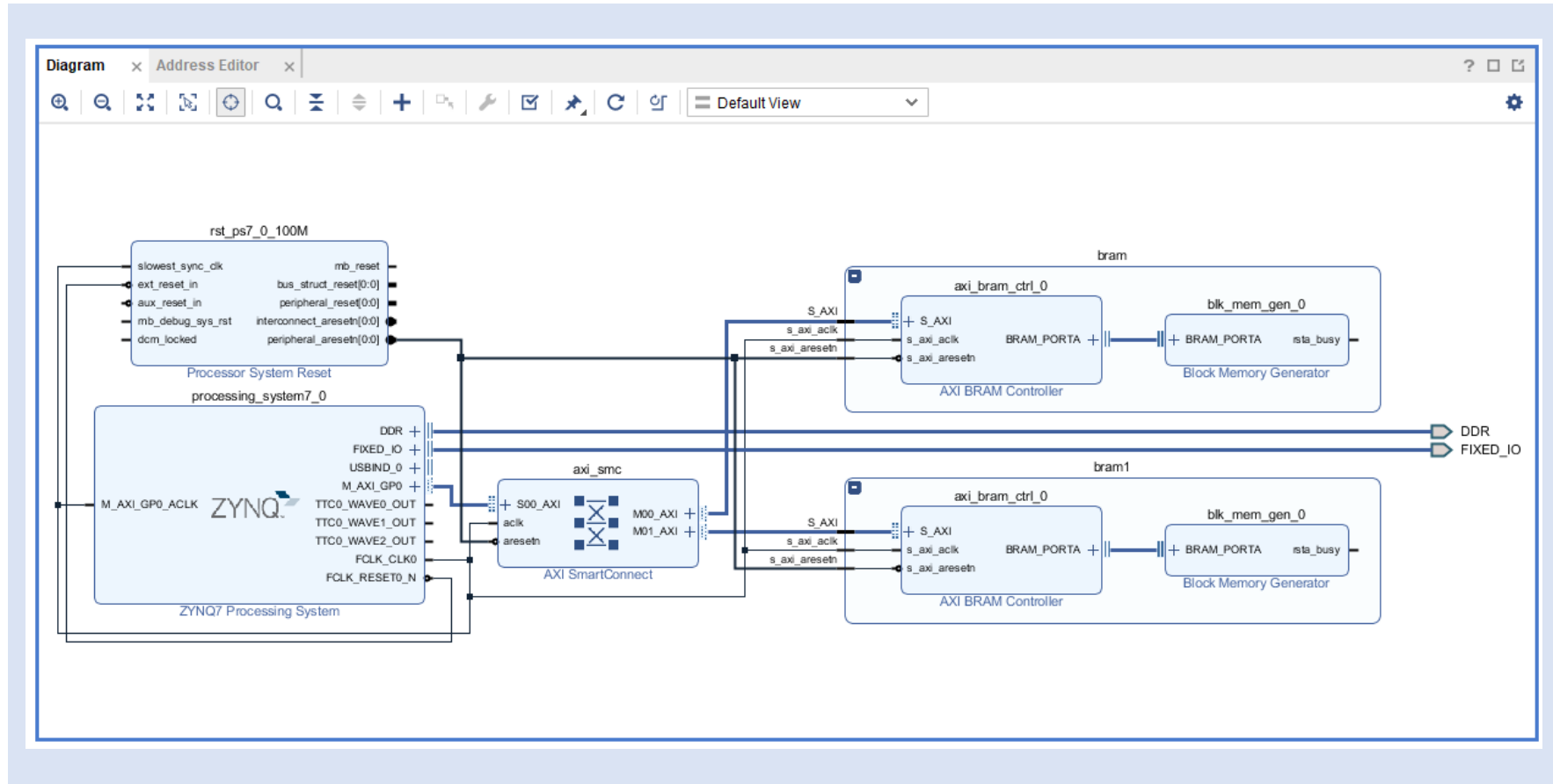
Lab 3: Overview and Introduction to Vivado

Step 17 – Run the connection automation and then check all options in the dialog and select OK



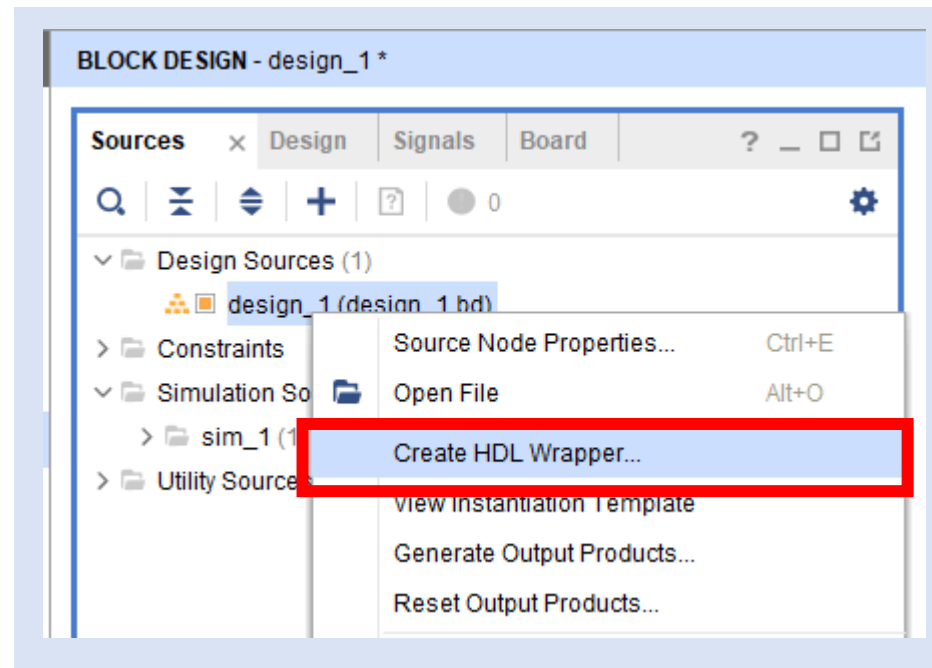
Lab 3: Overview and Introduction to Vivado

Step 18 – The completed diagram should look as below



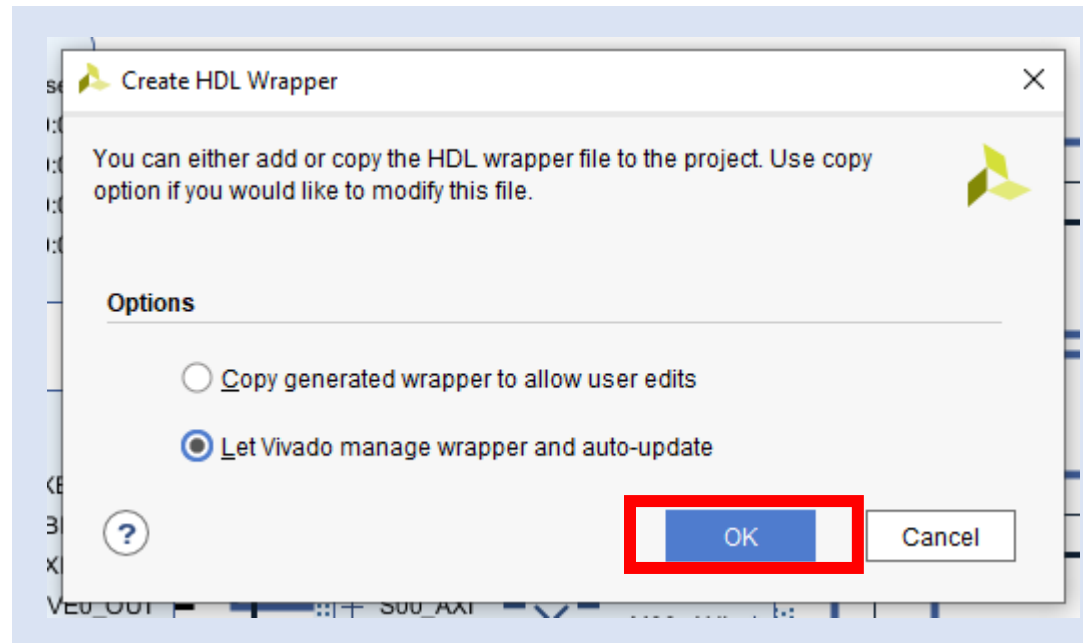
Lab 3: Overview and Introduction to Vivado

Step 19 – In the sources window, right click on the block diagram and select Create HDL Wrapper



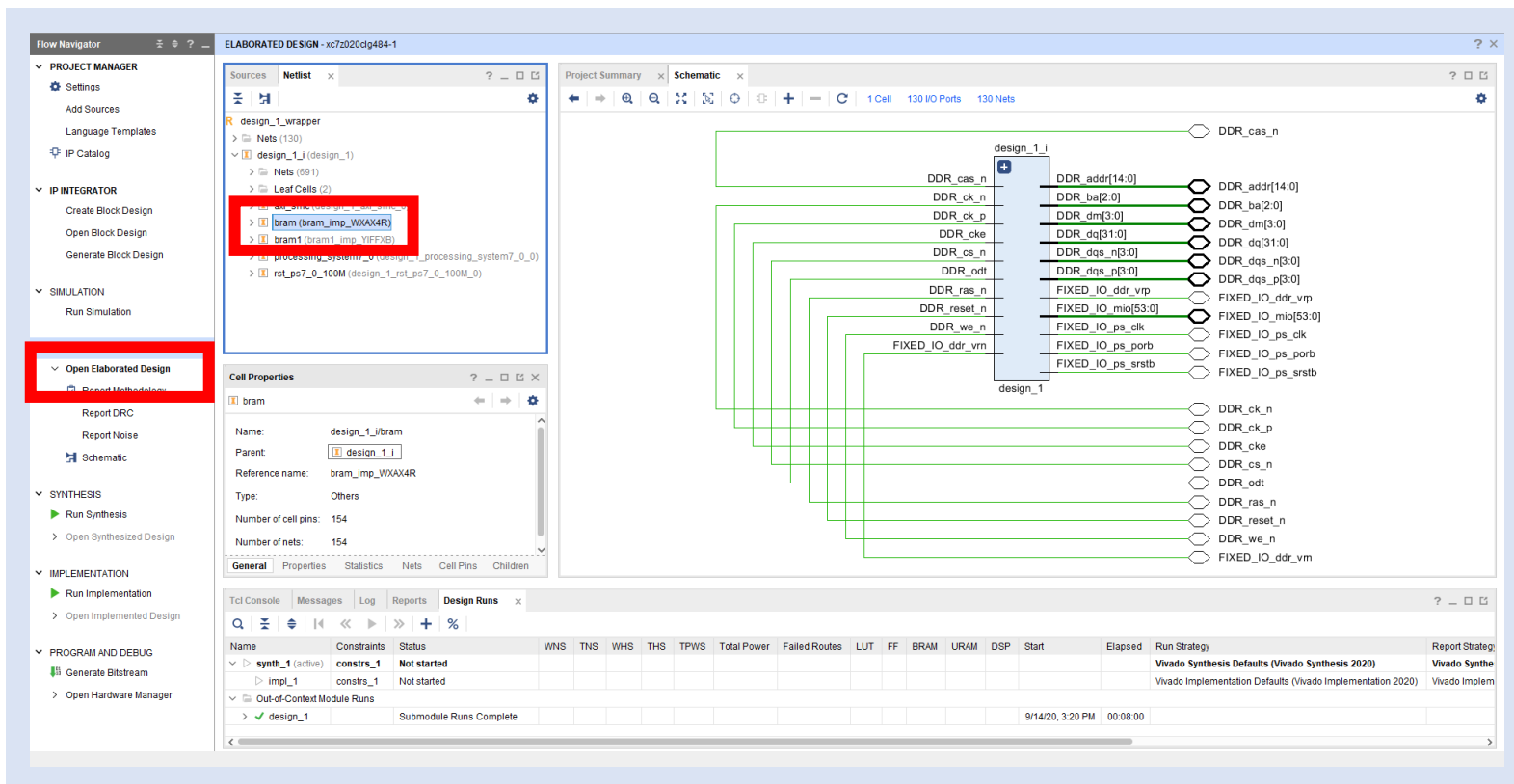
Lab 3: Overview and Introduction to Vivado

Step 20 – Let Vivado manage the HDL Wrapper



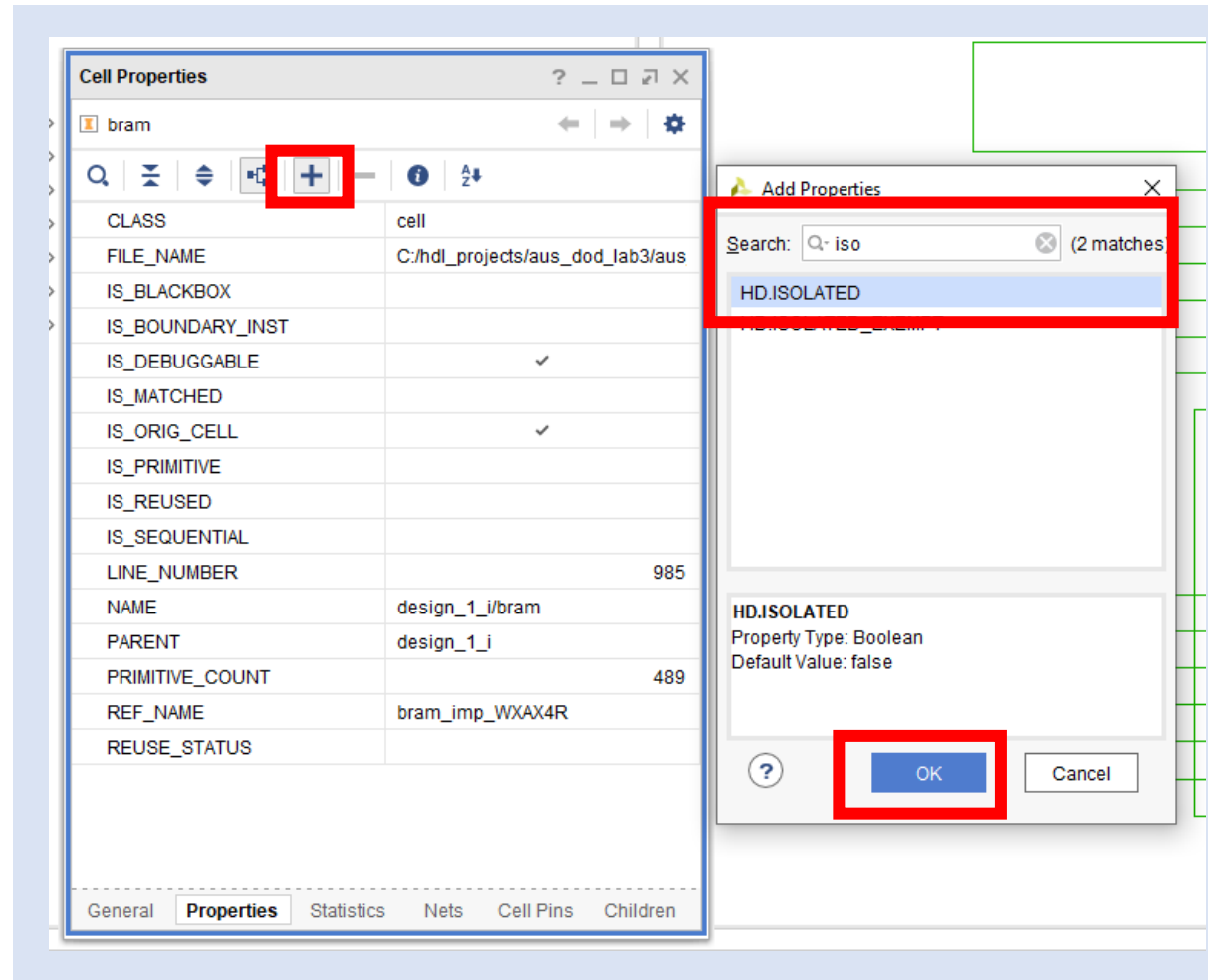
Lab 3: Overview and Introduction to Vivado

Step 21 – Open the elaborated design, this might take a few minutes – Select the BRAM and BRAM1



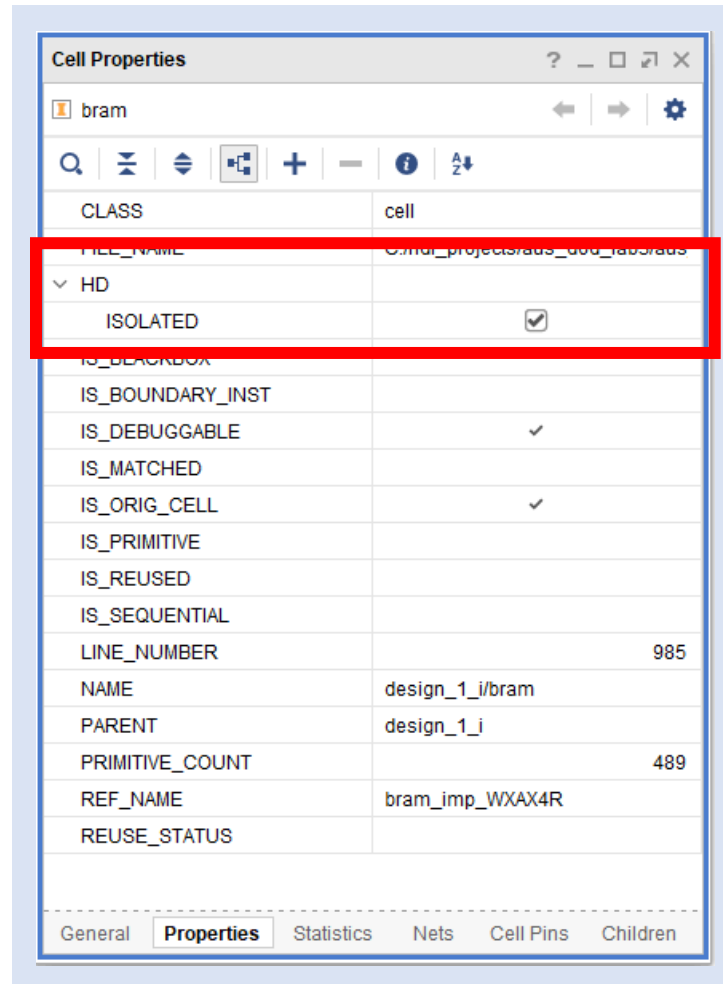
Lab 3: Overview and Introduction to Vivado

Step 22 – Click on Cell Properties and click on + enter isolated in the search bar – select HD Isolated



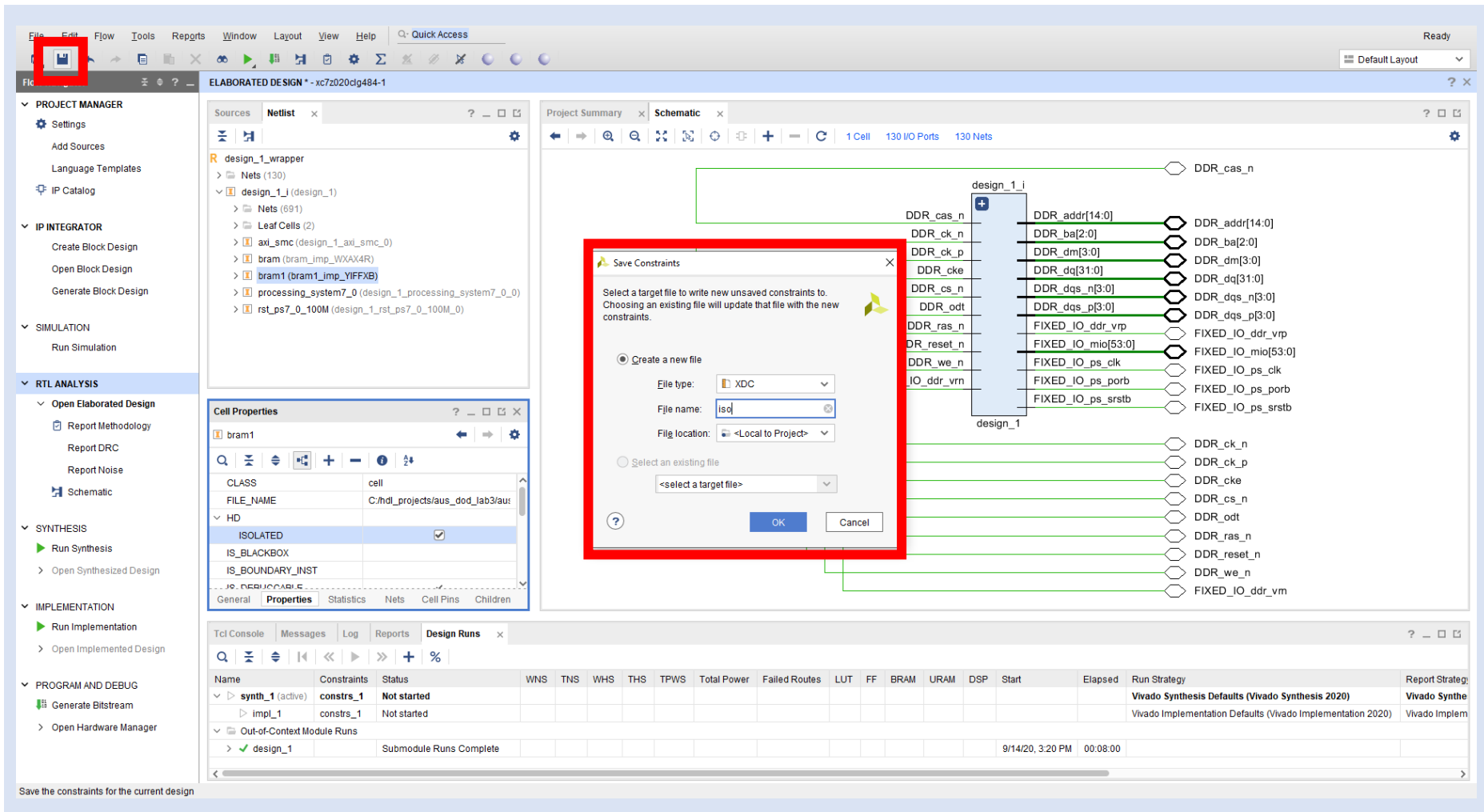
Lab 3: Overview and Introduction to Vivado

Step 23 – Double check the Isolated option is checked



Lab 3: Overview and Introduction to Vivado

Step 24 – Save the design and enter a name for the constraint file



Lab 3: Overview and Introduction to Vivado

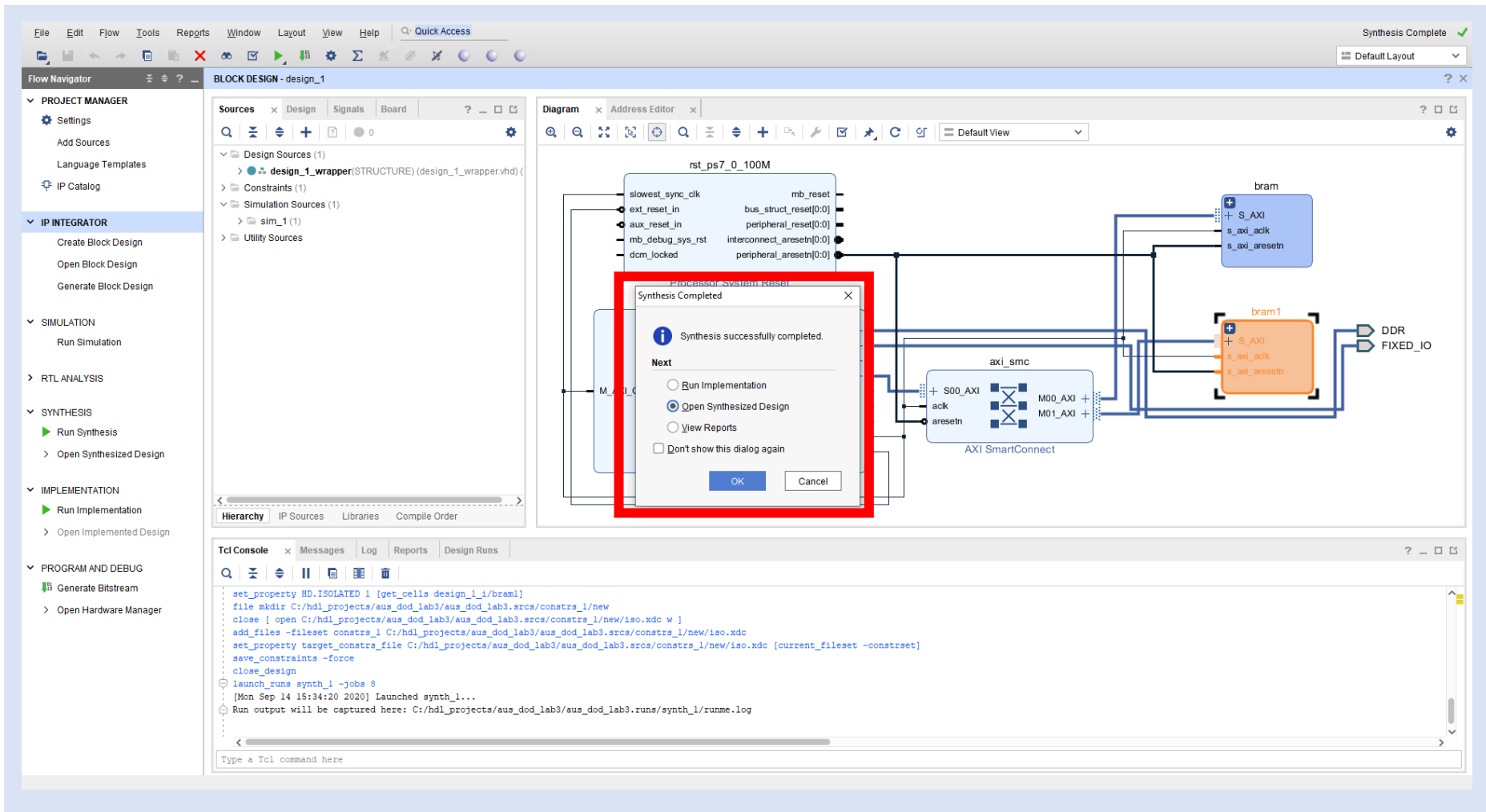
Step 25 – Run the synthesis

The screenshot displays the Vivado IDE interface for a project named 'design_1'. The left-hand 'Flow Navigator' pane shows the project workflow, with 'Run Synthesis' highlighted in a red box under the 'RTL ANALYSIS' section. The main workspace shows a block design diagram of a ZYNQ7 Processing System. Key components include the 'ZYNQ7 Processing System' block, an 'AXI SmartConnect' block, and two BRAM blocks labeled 'bram' and 'bram1'. The diagram illustrates the interconnection of these components, including clock signals like 'slowest_sync_clk' and 'M_AXI_GP0_ACLK', and reset signals like 'bus_struct_reset[0:0]' and 'peripheral_reset[0:0]'. The bottom pane shows the 'Tcl Console' with a series of commands for setting properties, creating directories, and launching the synthesis process.

```
set_property HD_ISOLATED 1 [get_cells design_1/bram1]
file mkdir C:/hdl_projects/aus_dod_lab3/aus_dod_lab3.srcs/constrs_1/new
close [ open C:/hdl_projects/aus_dod_lab3/aus_dod_lab3.srcs/constrs_1/new/iso.xdc w ]
add_files -fileset constrs_1 C:/hdl_projects/aus_dod_lab3/aus_dod_lab3.srcs/constrs_1/new/iso.xdc
set_property target_constrs_file C:/hdl_projects/aus_dod_lab3/aus_dod_lab3.srcs/constrs_1/new/iso.xdc [current_filesset -constrset]
save_constraints -force
close_design
launch_runs synth_1 -jobs 8
[Mon Sep 14 15:34:20 2020] Launched synth_1...
Run output will be captured here: C:/hdl_projects/aus_dod_lab3/aus_dod_lab3.runs/synth_1/runme.log
```

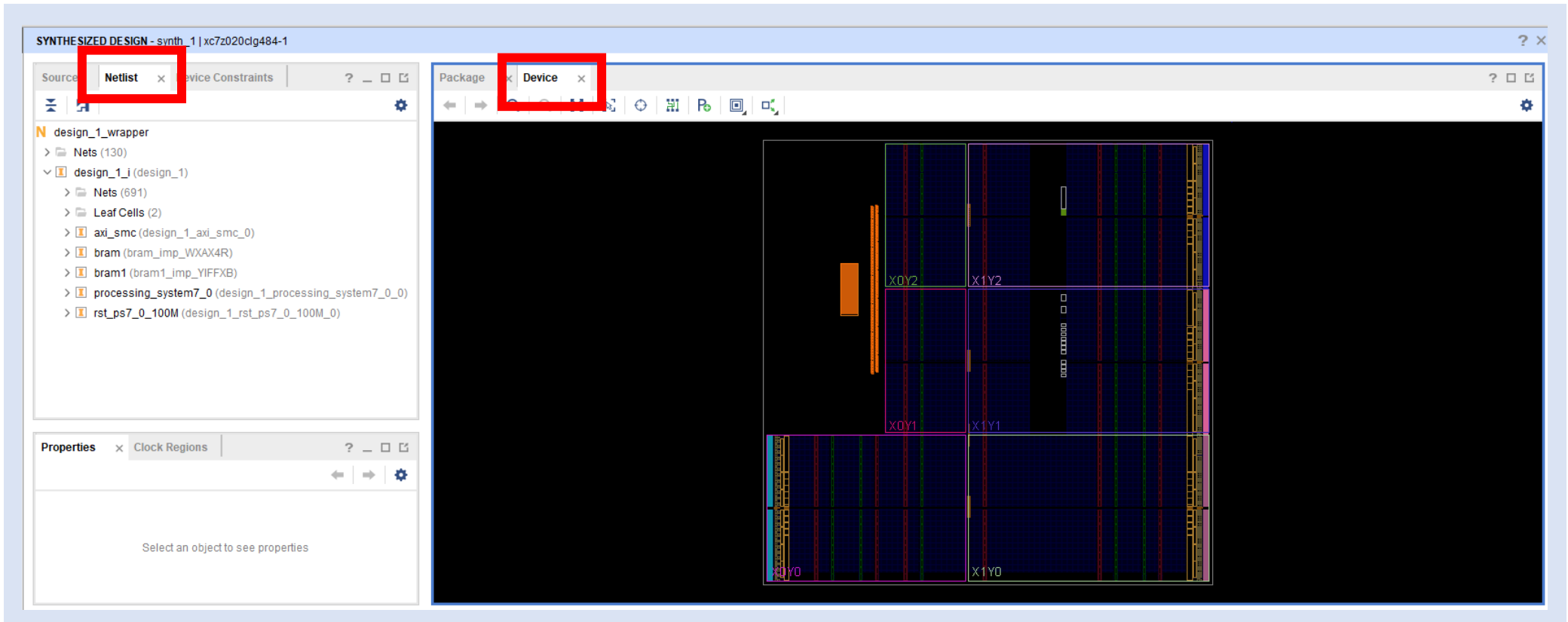

Lab 3: Overview and Introduction to Vivado

Step 26 – When the Synthesis is complete open the Synthesized design



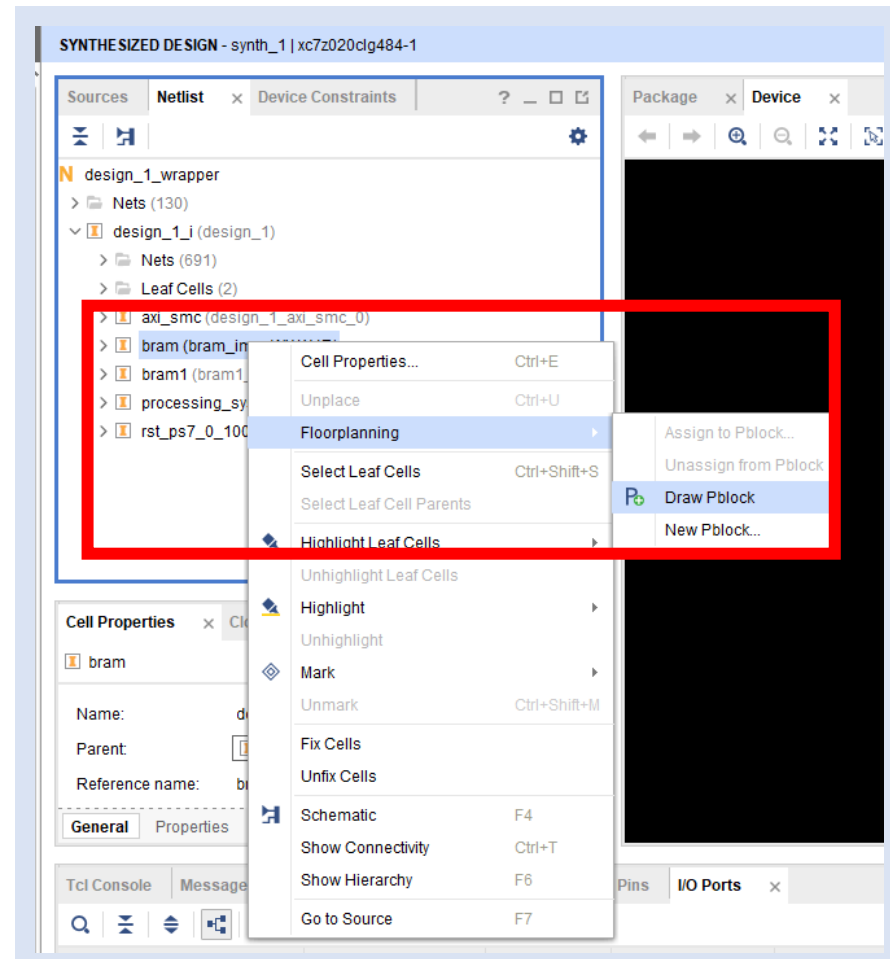
Lab 3: Overview and Introduction to Vivado

Step 27 – Click on the Netlist and Device Tabs once the synthesized view opens



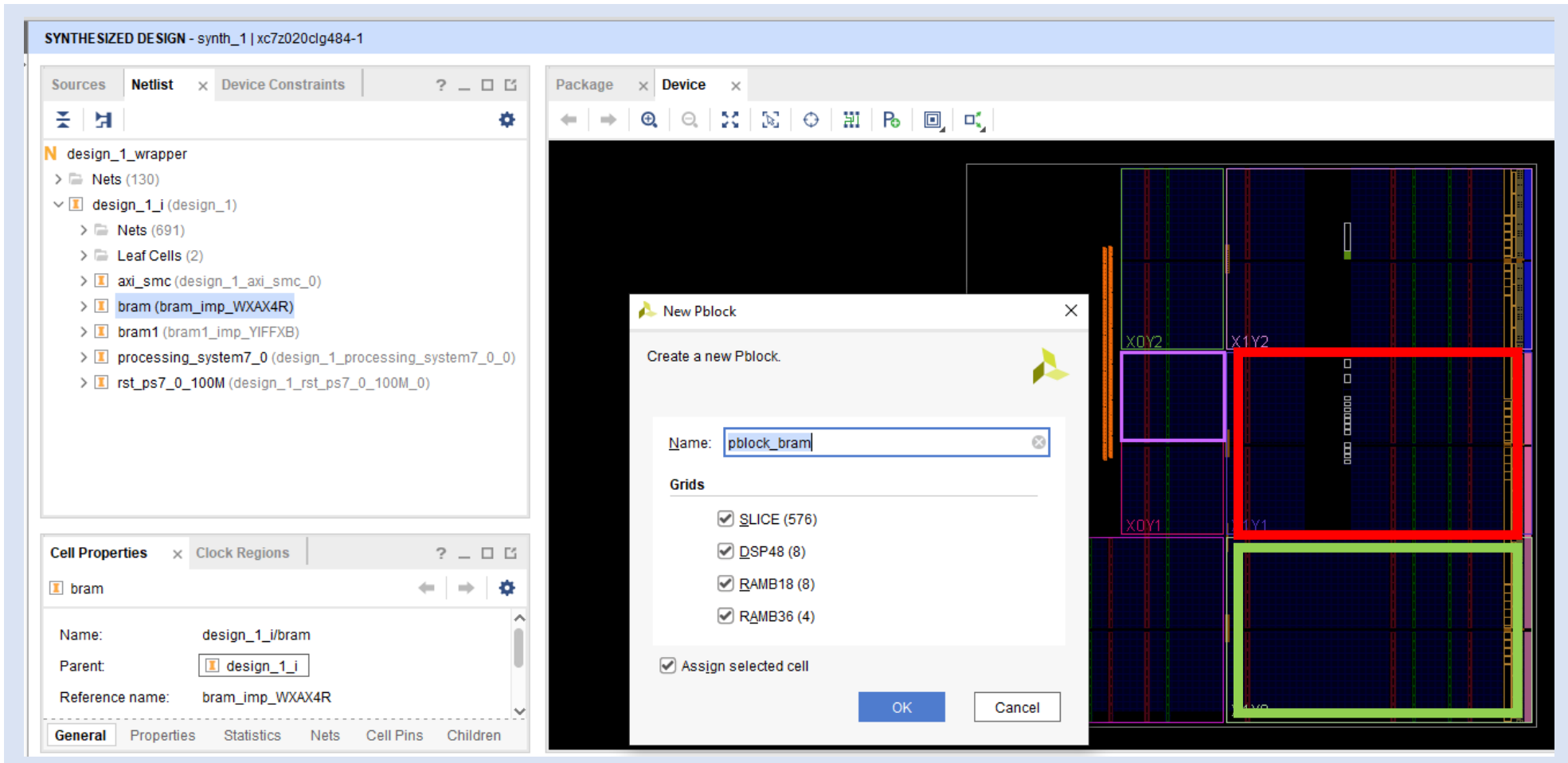
Lab 3: Overview and Introduction to Vivado

Step 28 – Right click on the BRAM and select Floorplanning create PBlock



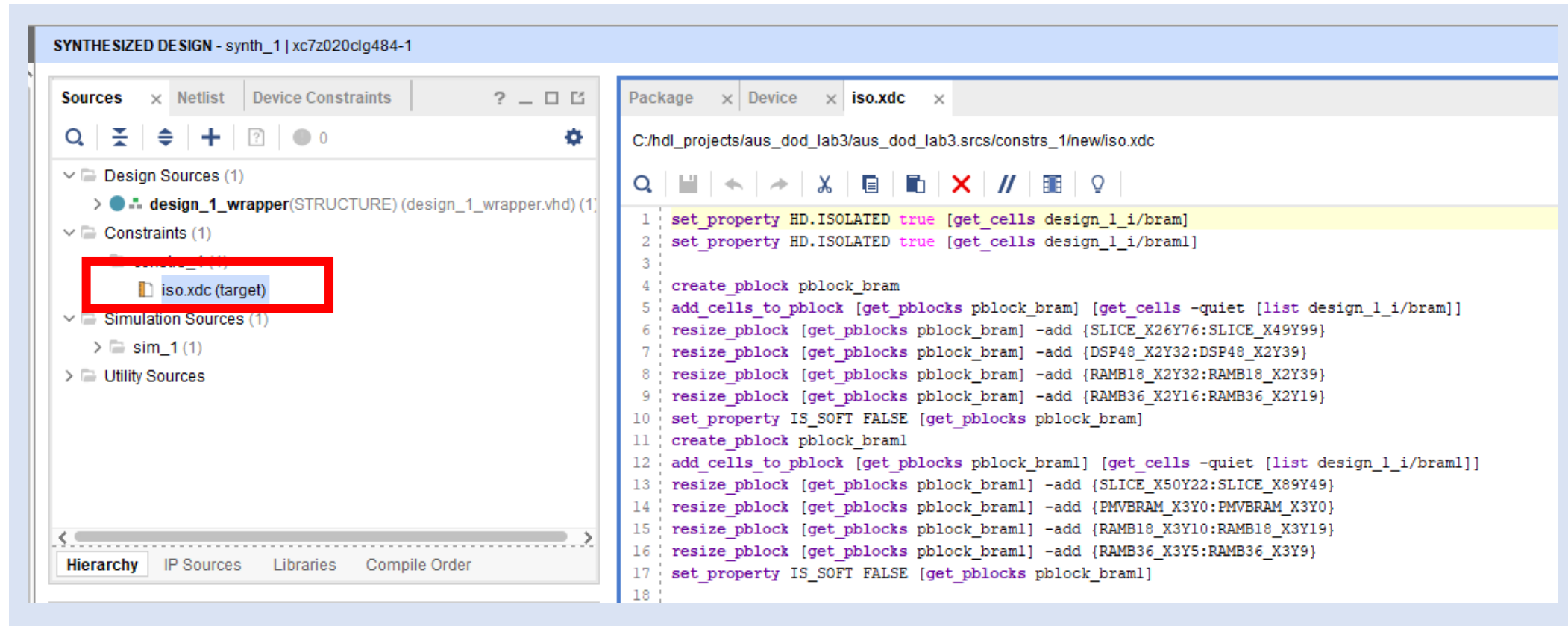
Lab 3: Overview and Introduction to Vivado

Step 29 – Select the area indicated in red and click OK to create the P Block. Do the same for the BRAM 1 in the area indicated in green



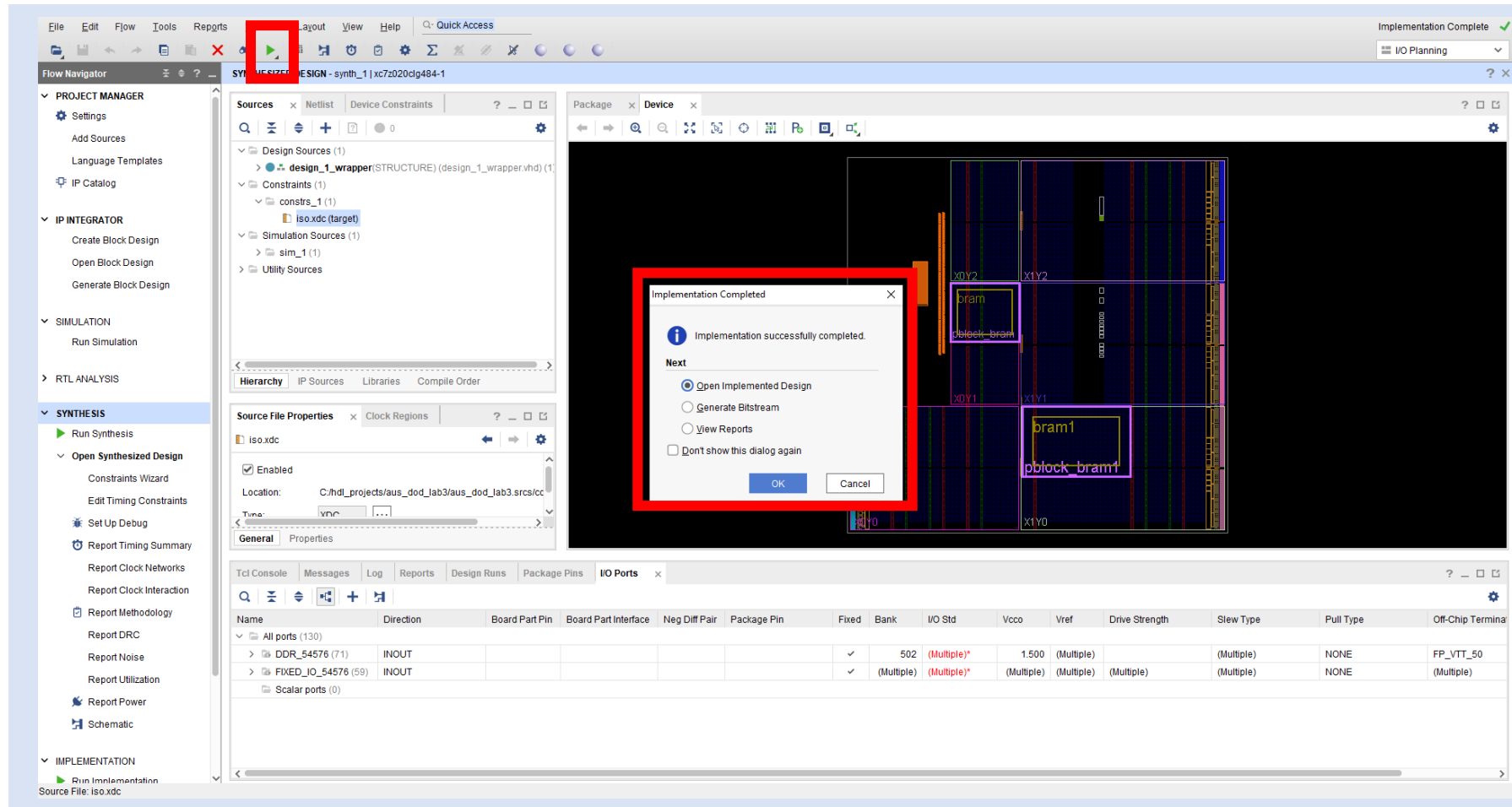
Lab 3: Overview and Introduction to Vivado

Step 30 – Open the XDC file and you will see the additional Pblock information added



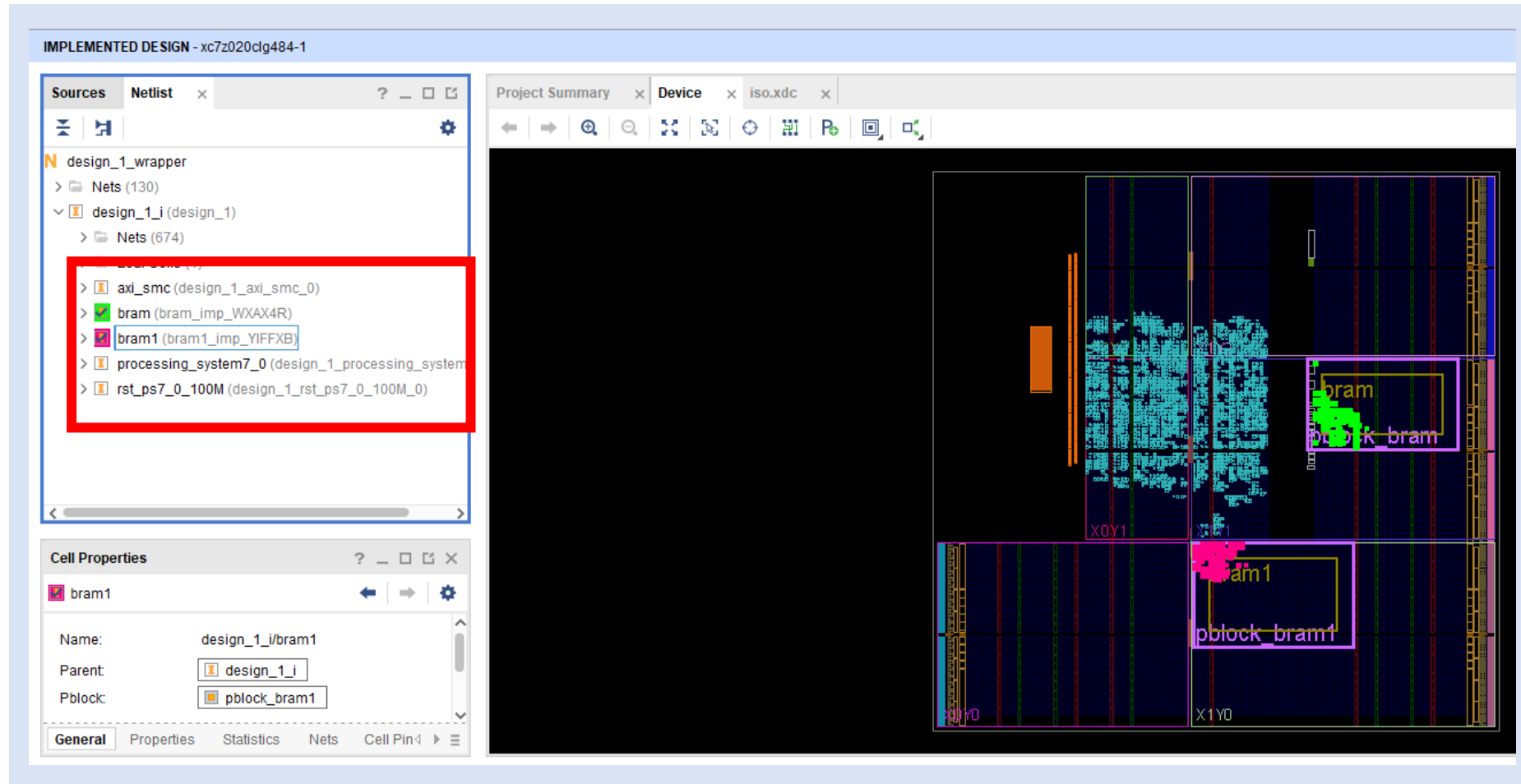
Lab 3: Overview and Introduction to Vivado

Step 31 – Implement the design and once the implementation is complete open the implemented design – if prompted close the synthesized design



Lab 3: Overview and Introduction to Vivado

Step 32 – Select the BRAM and BRAM1 in the netlist, right click and select color leaf cells. Select bright colors and you will see the BRAM isolated from each other



Lab 3: Overview and Introduction to Vivado

Step 33 – Select report DRC

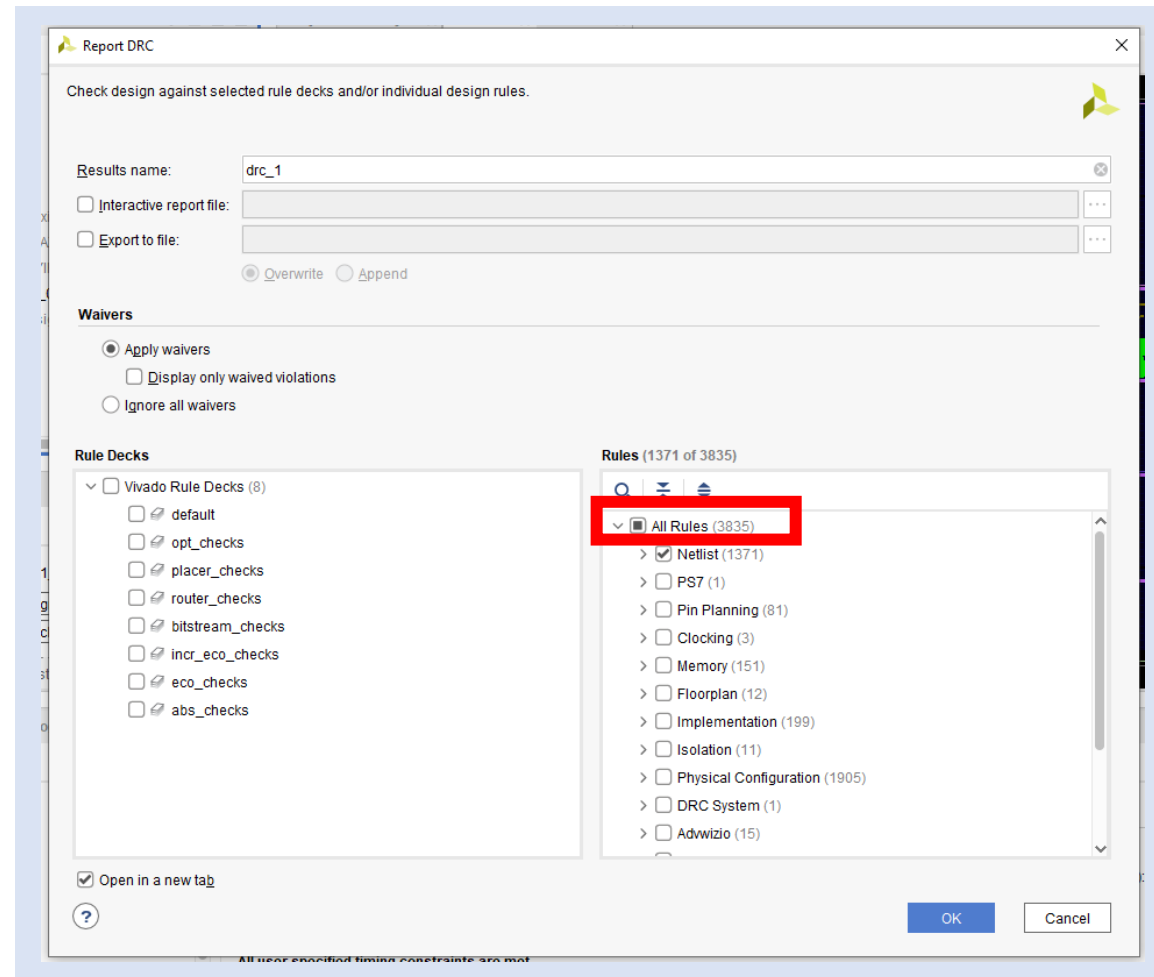
The screenshot displays the Vivado IDE interface. On the left, the 'Flow Navigator' pane shows the 'IMPLEMENTATION' section with 'Report DRC' highlighted in red. The main workspace is divided into several panes: 'Sources' (showing the project hierarchy), 'Cell Properties' (showing details for 'bram1'), 'Project Summary' (showing device and ISO files), and a large schematic view of the FPGA layout. At the bottom, the 'Timing' report is open, showing a 'Design Timing Summary' table.

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 2.978 ns	Worst Hold Slack (WHS): 0.013 ns	Worst Pulse Width Slack (WPWS): 3.750 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 18314	Total Number of Endpoints: 18314	Total Number of Endpoints: 6368

All user specified timing constraints are met.

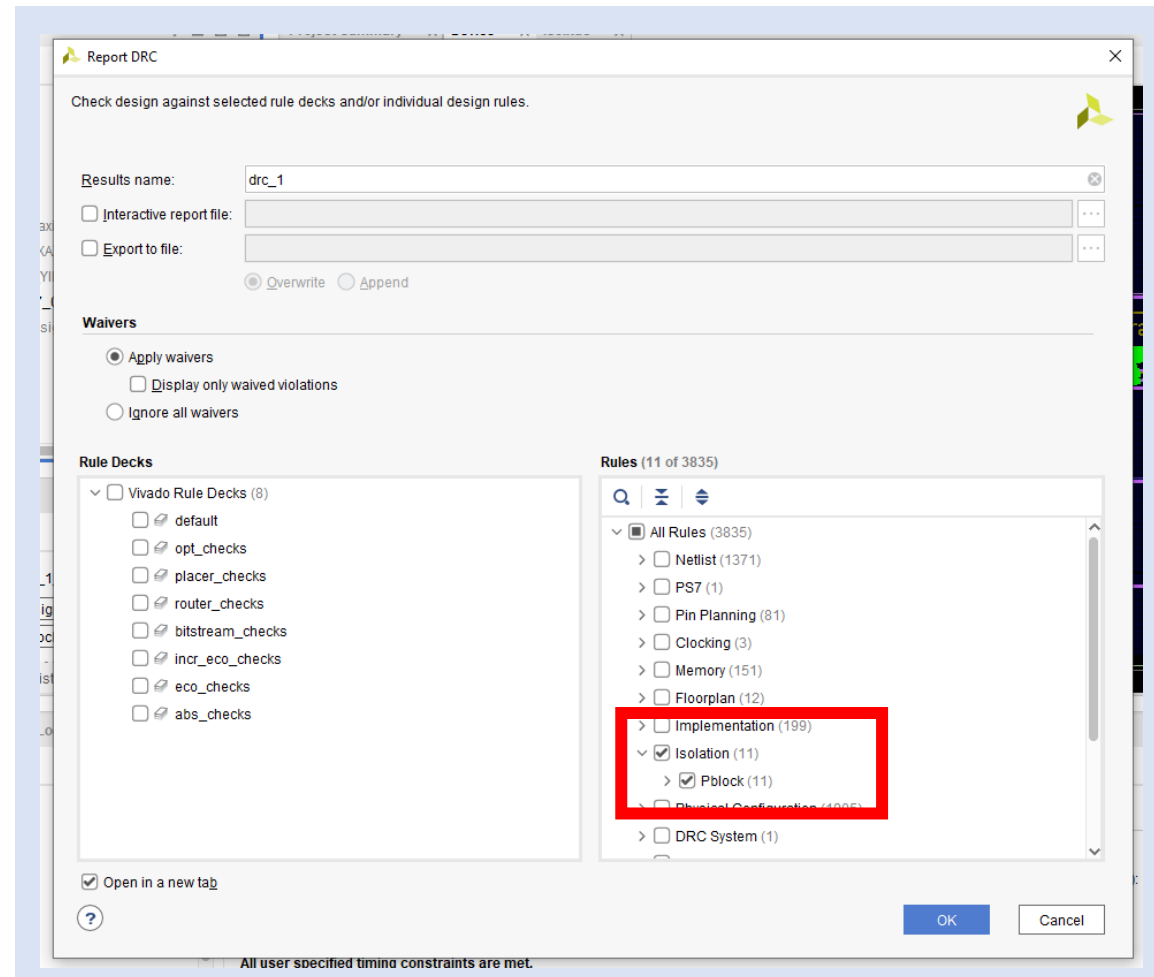
Lab 3: Overview and Introduction to Vivado

Step 34 – Uncheck All Rules



Lab 3: Overview and Introduction to Vivado

Step 35 – Enable Isolation Checks and click OK



Lab 3: Overview and Introduction to Vivado

Step 36 – No Violations should be found

The screenshot displays the Vivado IDE interface for a project named 'xc7z020clg484-1'. The 'Netlist' window on the left shows a hierarchical view of the design, including components like 'axi_smc', 'bram', and 'processing_system7_0'. The 'Cell Properties' window for 'bram1' is also visible. The main workspace shows a block diagram of the design. A 'Run DRC' dialog box is open in the center, displaying the message 'No Violations Found.' with an 'OK' button. The bottom panel shows the 'Design Timing Summary' window, which provides a detailed overview of the timing analysis results.

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 2.978 ns	Worst Hold Slack (WHS): 0.013 ns	Worst Pulse Width Slack (WPWS): 3.750 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 18314	Total Number of Endpoints: 18314	Total Number of Endpoints: 6368

All user specified timing constraints are met.